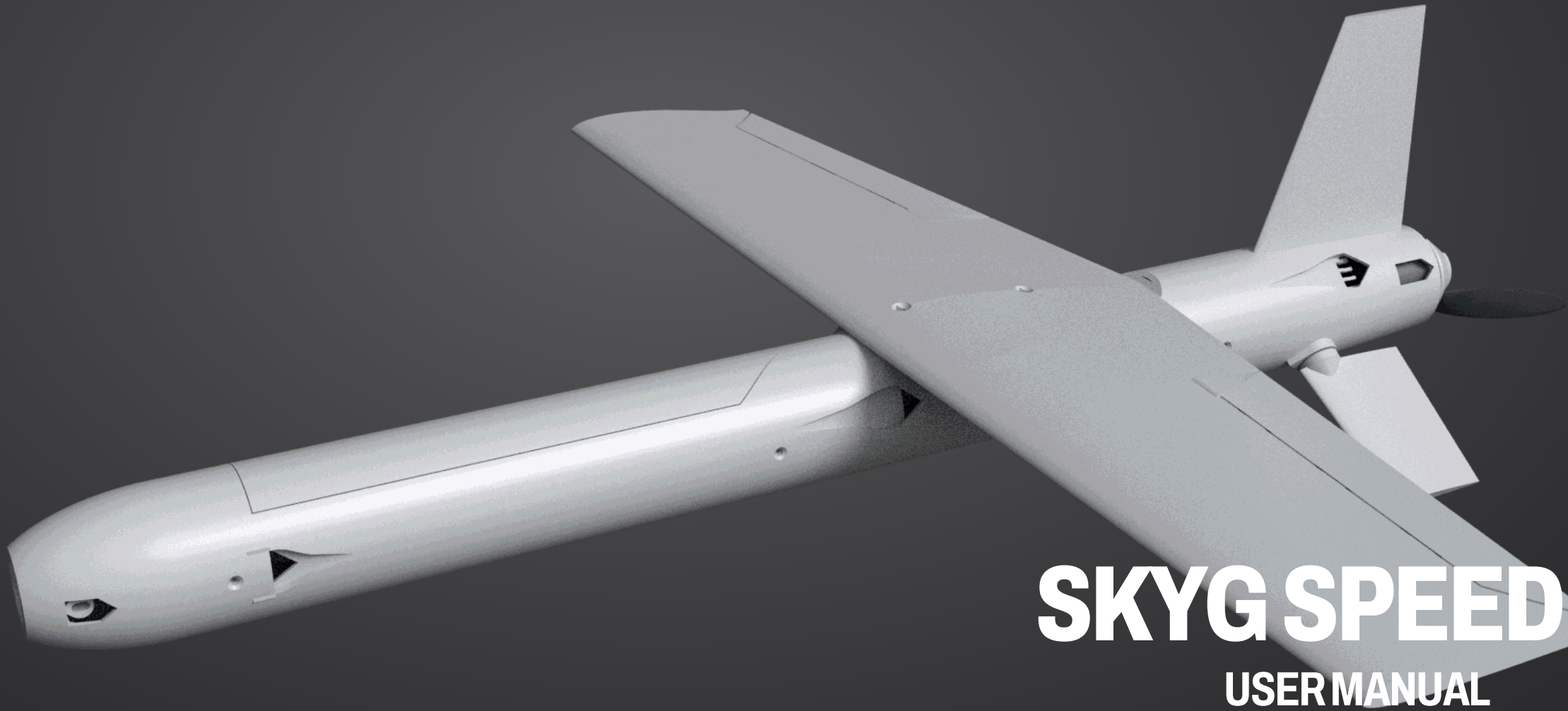


SKYG



SKYG SPEED 90

USER MANUAL

REVISION C6-9 · 2026-10-07

PACKAGE CONTENTS

Manual

SkyG Speed 90 user manual (this document).

STEP

Every printed part and the full assembly, for editing in any CAD program.

ArduPilot

Setup document and a starting .param file for the SpeedyBee F405 WING MINI.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

Wiring

The wiring diagram as a separate large page.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

TABLE OF CONTENTS

Introduction

General aircraft data	05
Aerodynamic design	06
Aerodynamic characteristics	07
Performance and speed stages	08
Compartment layout	10
Camera nose variants	11

Recommended equipment

Recommended electronics	13
Recommended accessories	14
Powertrain	15

Print settings

Print settings	17
Parts list	19

Parts orientation	20
-------------------	----

Assembly guide

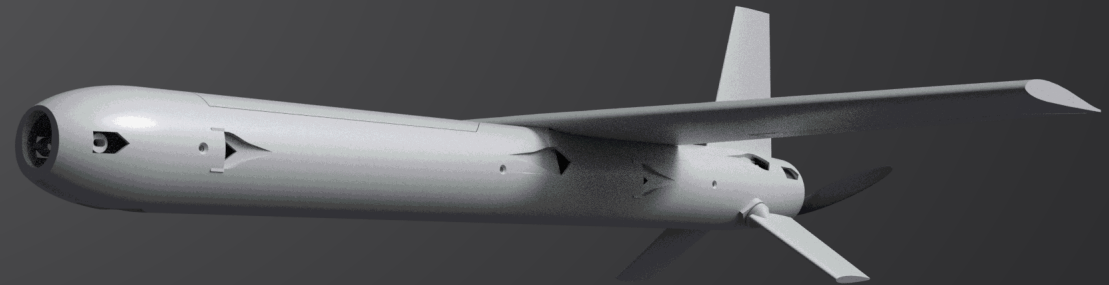
Fuselage assembly	28
Wing assembly	29
Tail module	32
Equipment and wiring	38
Camera nose	41
Final assembly	43
Finishing build	44
Wiring diagram	45
Electrical integration	46
ArduPilot configuration	49
Flight test and speed stages	50

SKYG SPEED 90

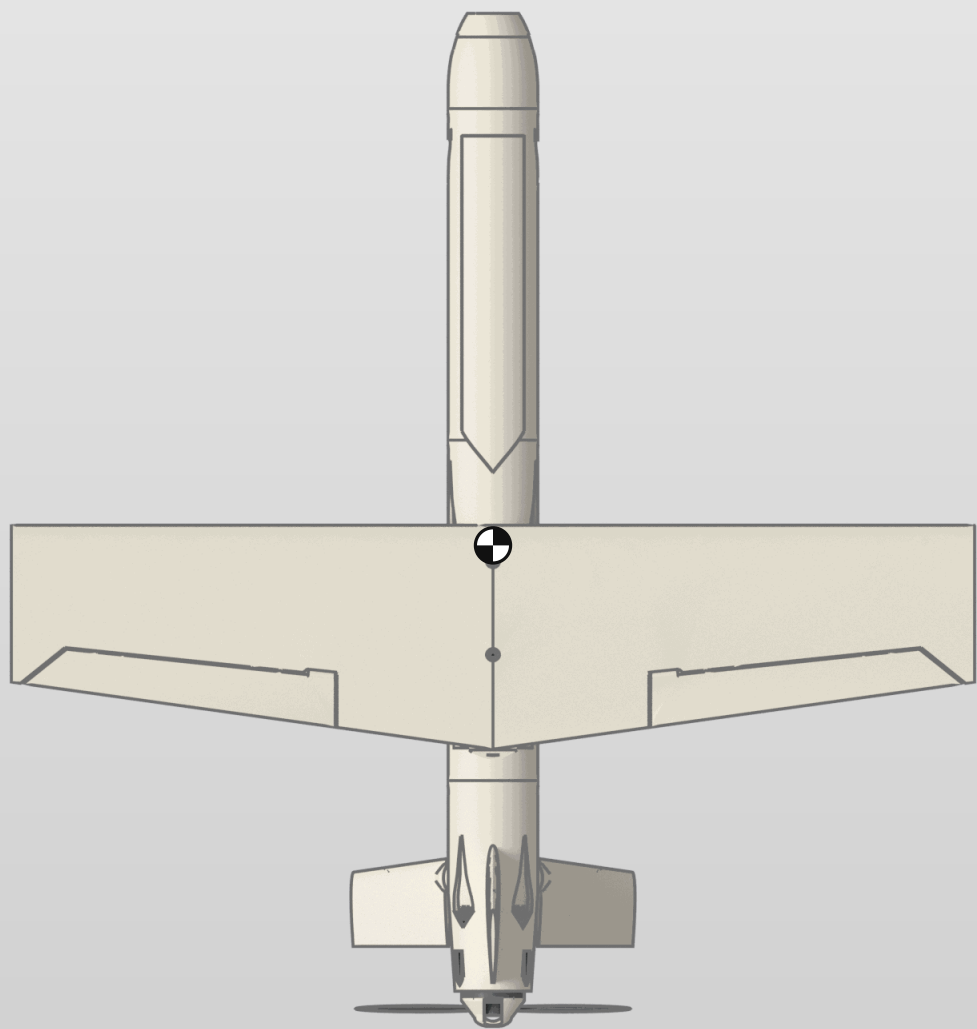
SkyG Speed 90 is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming PLA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 88 m/s finished with the planned rpm governor, and about 64 m/s until that governor is fitted, flown up through never-exceed stages.



GENERAL AIRCRAFT DATA



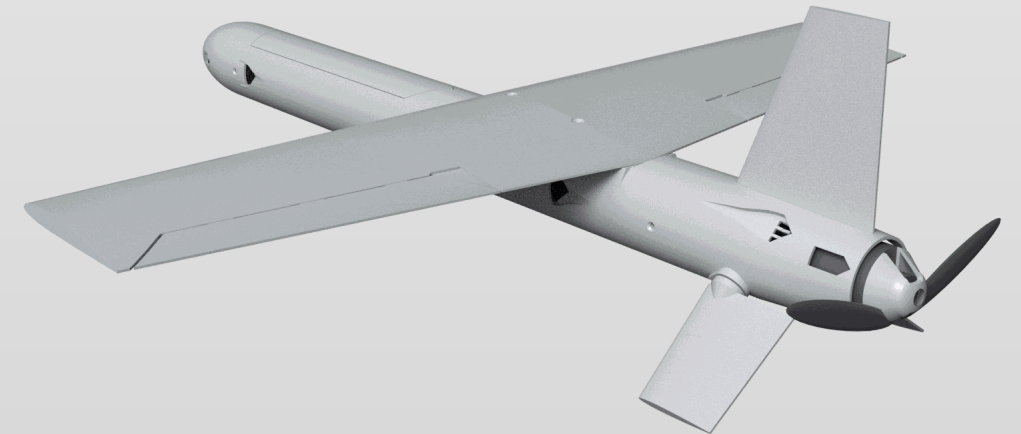
Wingspan	430 mm
Wing area	3.65 dm ²
Length	451 mm
Centre of gravity	9 mm behind the wing leading edge at the root
All-up weight	390 g (micro-camera nose)
Stall speed	13.0 m/s
Top speed (model)	87–88 m/s with the governor
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	104 g/dm ²

AERODYNAMIC DESIGN

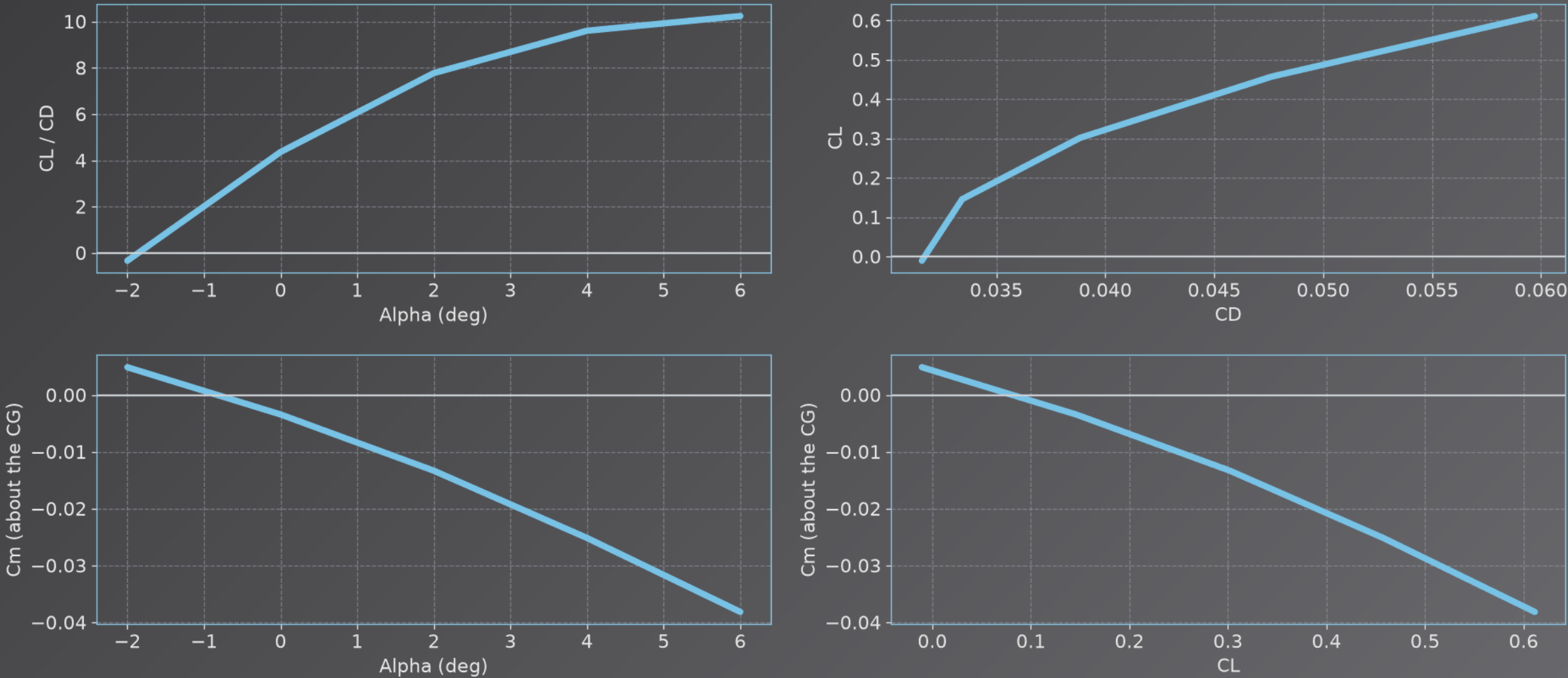
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 12 % of the mean chord, 6.5 % static margin ahead of the neutral point (x 252 mm from the nose datum): stable without a tail download penalty at speed.

Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 122 m/s, above 1.2 × the 90 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.



AERODYNAMIC CHARACTERISTICS



Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 10.2 at 6.0°; C_m about the design CG.

PERFORMANCE AND SPEED STAGES

Motor	2207 2550KV 4S racing motor
Propeller	APC 5x7.5E (LP05075E)
Battery	Tattu R-Line 4S 650 75C
Static thrust	1064 g at 51 A (thrust / weight 2.72)
Throttle cap (prop rpm rating)	70 %
Top speed	63–64 m/s now; 87–88 m/s with the governor
Full-throttle time	about 0.7 min per pack (bursts of ≤ 10 s)

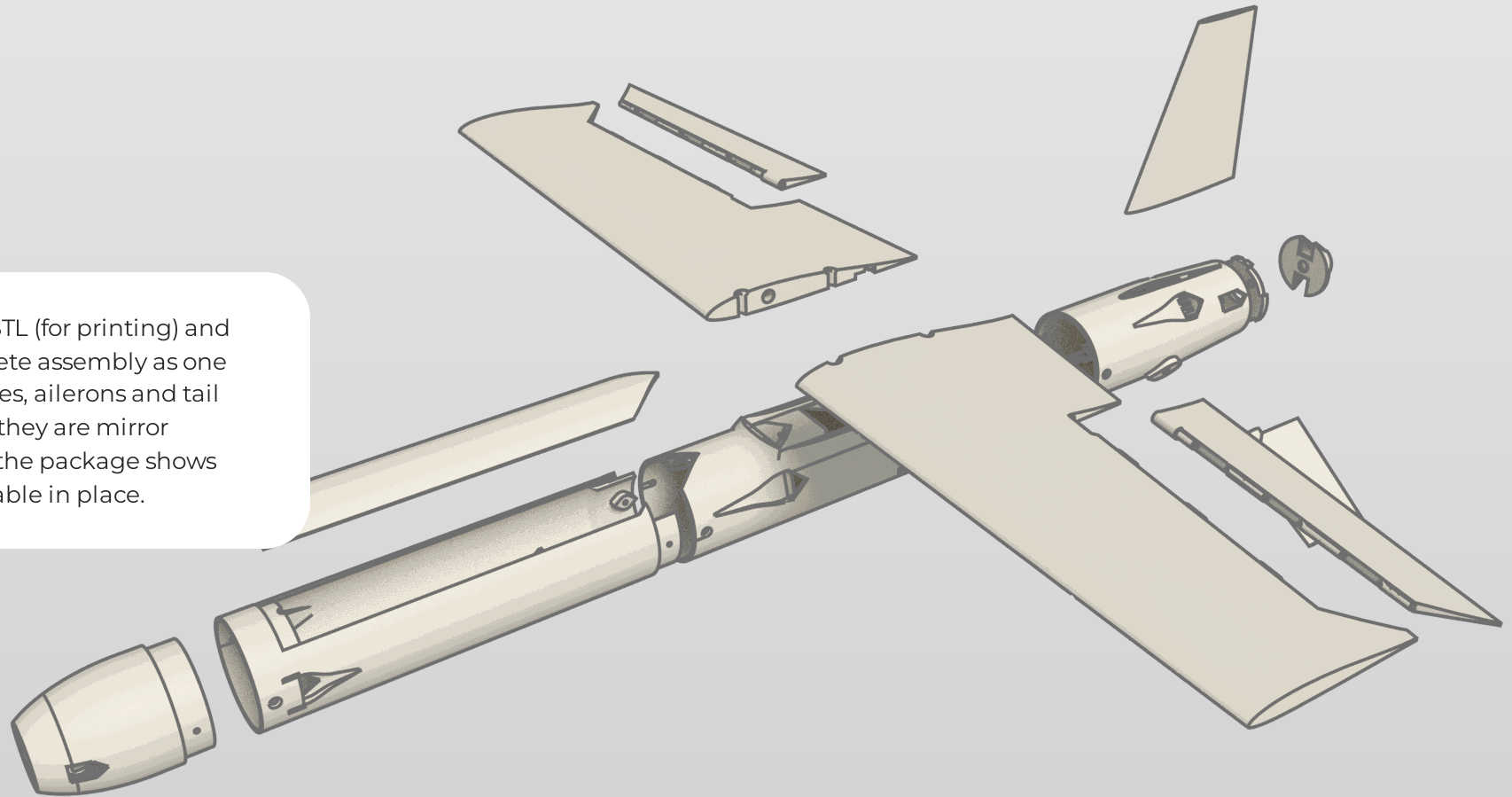
Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	27 %	24 m/s
35 m/s	32 %	29 m/s
40 m/s	36 %	33 m/s
45 m/s	41 %	38 m/s
50 m/s	45 %	41 m/s
55 m/s	50 %	46 m/s
60 m/s	54 %	49 m/s
65 m/s	59 %	54 m/s
70 m/s	64 %	58 m/s
75 m/s	69 %	62 m/s
80 m/s	70 % (needs the governor)	63 m/s
85 m/s	70 % (needs the governor)	63 m/s
90 m/s	70 % (needs the governor)	63 m/s

No airspeed sensor: each stage is held by a throttle limit on a transmitter switch (LAUNCH = full cap for the throw and climb-out only, STAGE = the limit). Level runs only; speed from GPS two-way passes.

FILES



Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



COMPARTMENT LAYOUT

BATTERY BAY

4S 650 pack, 60 × 31 × 21 mm max; slides x 98–130 for balance; rear stop at x 160

WING BAY

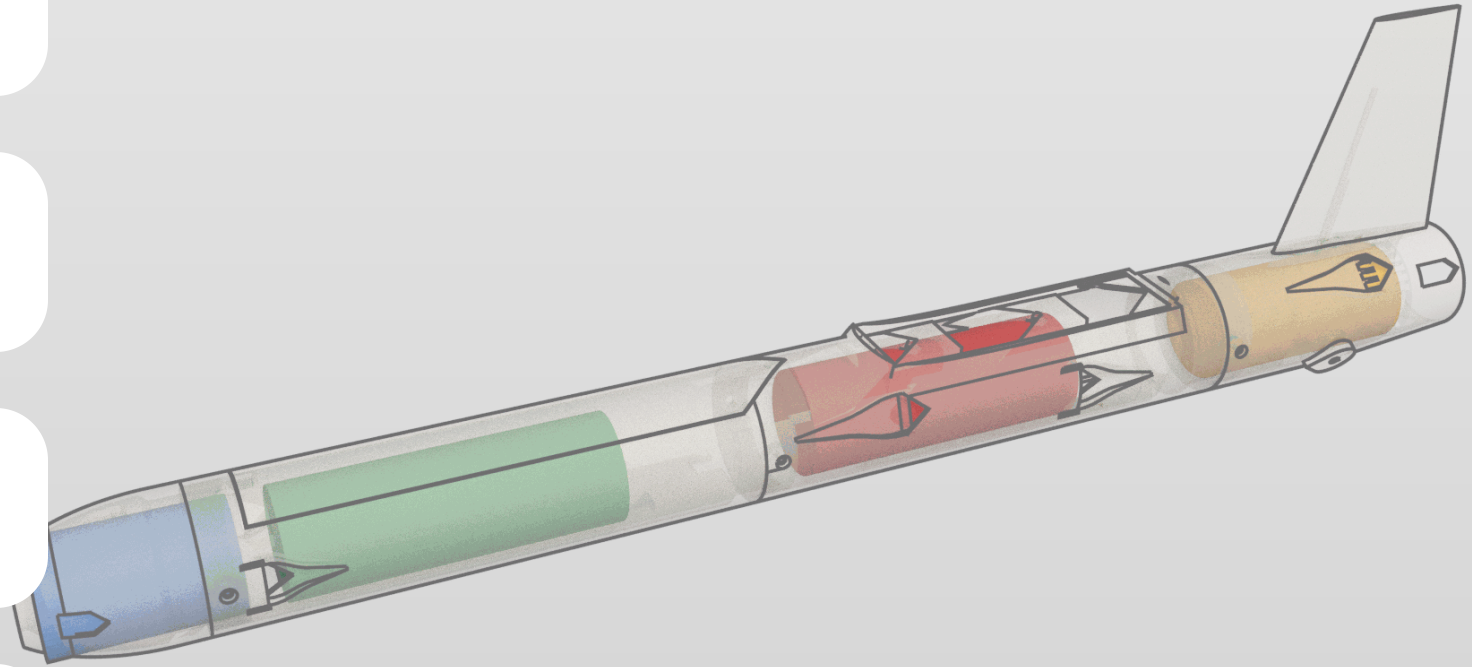
Flight controller stack with the ESC on top, receiver, aileron servo; reached with the wing off

TAIL MODULE

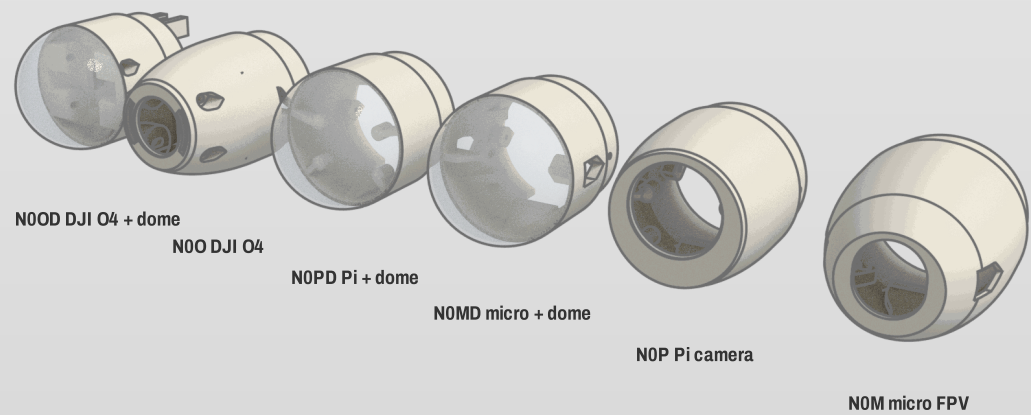
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

NOSE BAY

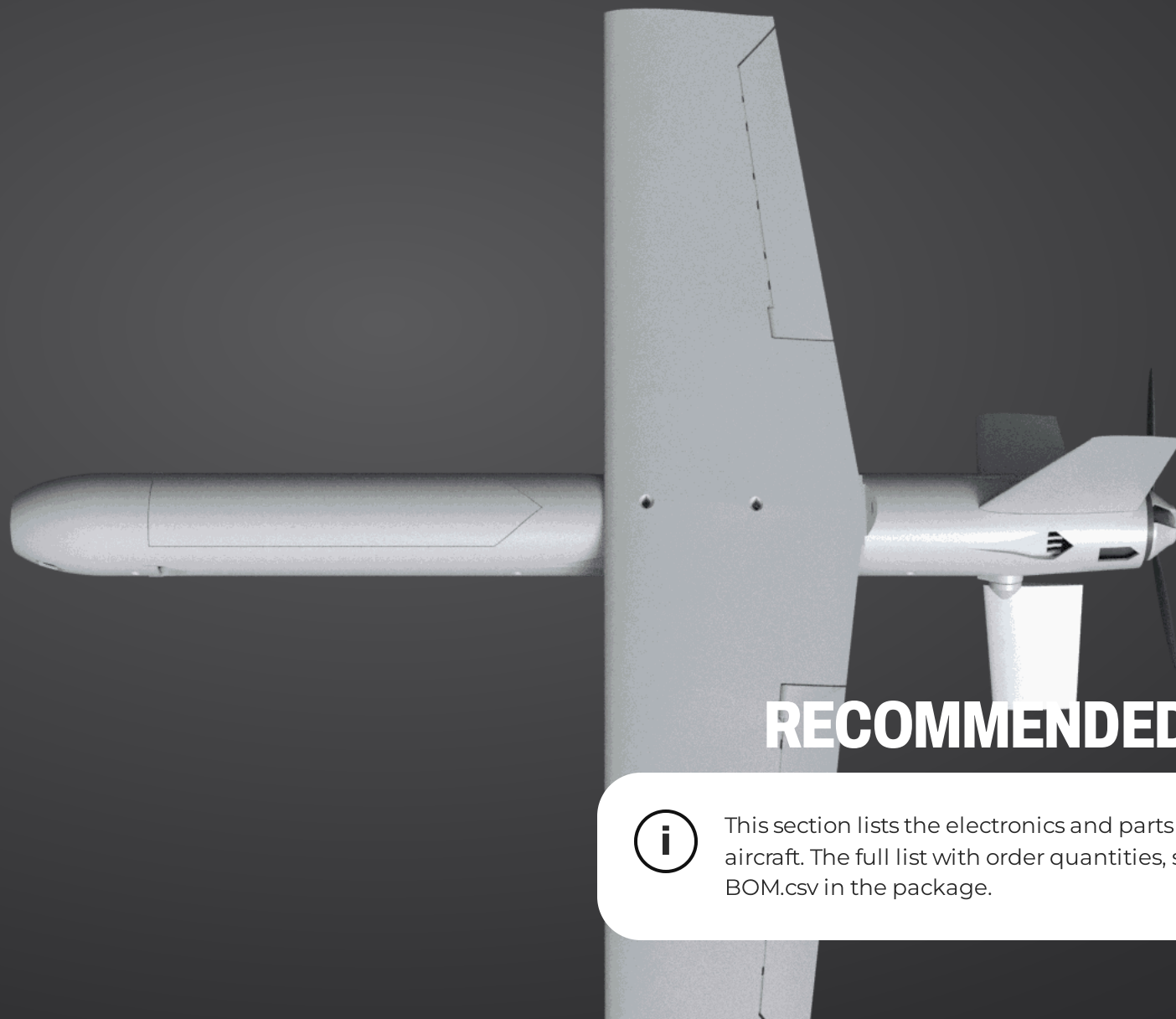
Camera and VTX (or the DJI O4 unit) in the removable nose



CAMERA NOSE VARIANTS



Nose	Camera	Battery centre for balance
NOM	19 mm micro FPV camera + 20 × 20 VTX	x 115 mm
NOP	Raspberry Pi Camera Module 3 (test option)	x 94 mm (at its stop + nose ballast)
N0MD	micro camera behind a 40 mm dome, nano VTX under the hatch	x 114 mm
N0PD	Pi camera behind a dome	x 99 mm
N0O	DJI O4 Air Unit (Lite), ducted	x 119 mm
N0OD	DJI O4 behind a dome	x 121 mm



RECOMMENDED EQUIPMENT



This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT	MODEL / SPECIFICATION
Motor	2207 2550KV 4S racing motor
Propeller (fixed)	APC 5x7.5E (LP05075E)
ESC	HDZero Halo 4in1 70A ESC AM32
Flight controller	SpeedyBee F405 WING MINI
GPS + compass	TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos	3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery	Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g)
Receiver	RadioMaster RP1 V2
FPV camera	Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna	20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)	DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
XT30 connector pairs	1
Carbon tube	1
Music wire 2.0 mm	2
Pushrods 1.2 mm with Z-bend	4
Pushrod keepers for 1.2 mm wire	4
Z-bend pliers, 1.1 mm drill, 1.2 mm drill (as a hand reamer), small flat file	0
Nylon countersunk screw M3 x 16	1
M3 hex nut	2
M2 x 6 countersunk screws (joints)	6
Magnets 3x2 mm	2
Magnets 3x1 mm	2
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	1

ITEM	QTY
Heat shrink + silicone wire	as needed
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 100 mm	2
Motor screws M3	4
Glass skin + finish	14 g
Nylon countersunk screw M3 x 12	1
M2 x 5 screws	4
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
Vent tape	1
15T servo arms	1
Dupont housings + crimps (1-pin, 4-pin)	3

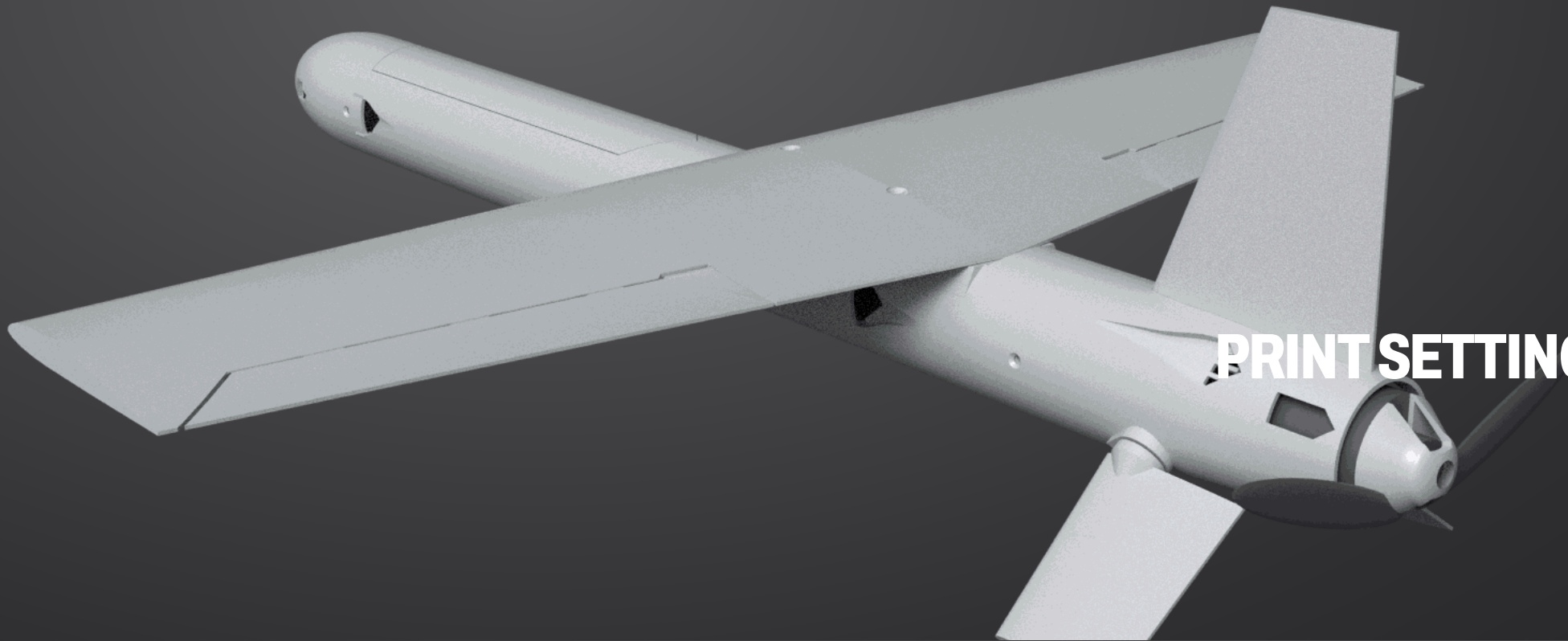


Motor and prop. 2207 2550 KV class 4S racing motor (e.g. 2207 2550KV: 27.9-28 mm bell, about 21.5 mm body, 33.5 mm with the 5 mm prop shaft, 33 g, 16 x 16 mm M3). Prop: APC 5 x 7.5E fixed thin-electric pusher (LP05075E, 5 mm bore, hub 12.7 x 7.1 mm, 30,000 rpm rating), rpm-governed at 29,500.

Throttle cap. A 5-inch prop is rated about 30,000 rpm. On a full 4S pack the unloaded prop would exceed it at full throttle, so the flight controller caps the throttle at 70 %. Never raise it, and inspect the blades after every thrust-stand run and flying day.

rpm governor (open item). This version is designed around a governor that holds the prop at 29,500 rpm, which lets it use the full rated rpm in level flight. ArduPilot's governor script does not run on the F405 flight controller, so until an H7 flight controller or a governing ESC is fitted the throttle cap above applies and the top speed is about 64 m/s; the 80–90 m/s stages need the governor.

Battery. Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g); XT30. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1064 g at 51 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS



The airframe is designed for **Bambu PLA Aero** (foaming, lightweight) on a **0.4 mm nozzle** and **0.20 mm layers**; every part fits a Bambu H2S or H2D build volume. The tail module may be printed in ASA Aero, the motor plate and spinner in PETG-CF.

GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, 30 % lightning infill (it only grows under top surfaces: at 0 % the solid bosses and bridges print hollow and their tops collapse).
- Wing halves and ailerons: 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Motor plate and spinner: PETG-CF, 4 walls, 40 %.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown, and checked on the slicer's own G-code with these settings: nothing is printed over air and no bridge is longer than 10 mm. Keep supports switched off and use the supplied .3mf files (they carry the settings). Check the top of the aileron's torque-rod bore on the first aileron.

PRINT SETTINGS

FIRST PART AND WEIGHT CHECK

Print F1 first and weigh it. If it is more than 15 % over the sliced mass (about 17 g), lower the PLA Aero temperature or flow before printing the rest; foaming filaments vary with temperature, humidity and the enclosure.

COUPON TESTS

Before the full build, print the last 25 mm of F1 with the first 25 mm of F2, and the F2/F3 and nose joints the same way: every joint must slide on and off by hand and the M2 screws must bite. Print a 40 mm wing-root section and check the spar fit.

SLICER

The masses in this manual come from Bambu Studio with the H2S 0.4 mm profile and the walls/infill listed on the orientation pages. Import the STL files in millimetres; keep each part in the orientation shown.

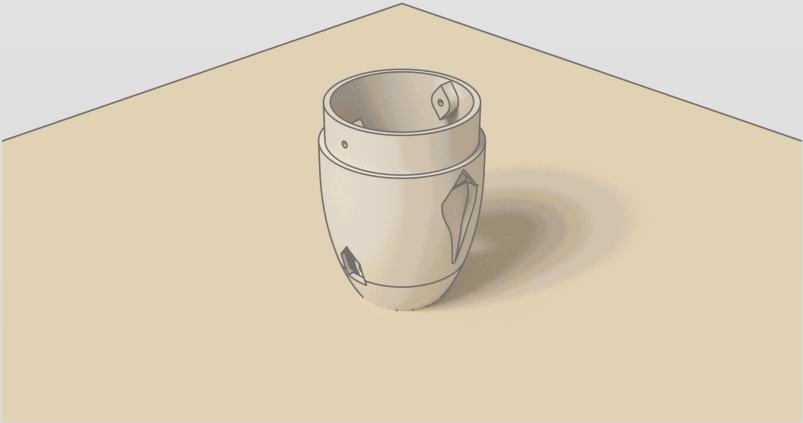
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	PLA Aero	6.2 g
N0P NOSE (PI CAMERA)	PLA Aero	5.9 g
N0MD NOSE (MICRO + DOME)	PLA Aero	3.9 g
N0PD NOSE (PI + DOME)	PLA Aero	4.3 g
N0O NOSE (DJI O4)	PLA Aero	8.5 g
N0OD NOSE (DJI O4 + DOME)	PLA Aero	6.7 g
F1 FUSELAGE FRONT	PLA Aero	17.4 g
F2 FUSELAGE MAIN	PLA Aero	21.7 g
F3 TAIL MODULE	ASA Aero	7.4 g
HF FRONT HATCH	PLA Aero	3.8 g
FN FIN	PLA Aero	4.5 g
VTX SHELF (N0M)	PLA Aero	0.4 g
DOMERING	PLA Aero	0.5 g
HG GPS CARRIER	PLA Aero	0.9 g

PART	MATERIAL	MASS
WR WING HALF RIGHT	PLA Aero	16.3 g
WL WING HALF LEFT	PLA Aero	16.3 g
ARAILERON RIGHT	PLA Aero	2.2 g
ALAILERON LEFT	PLA Aero	2.3 g
SR TAIL PANEL RIGHT	PLA Aero	3.0 g
SL TAIL PANEL LEFT	PLA Aero	2.9 g
M1 MOTOR PLATE	PETG-CF	1.9 g
SP SPINNER	PETG-CF	1.7 g

One aircraft with the N0M nose: 110 g printed. Only one camera nose is needed; the others are options.

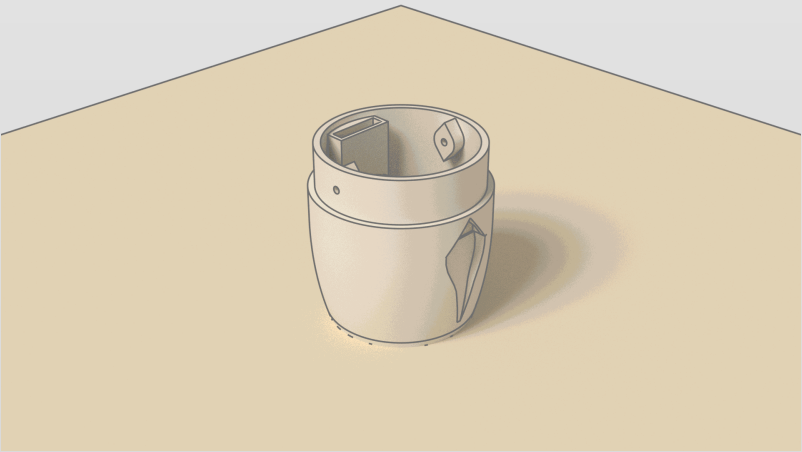
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

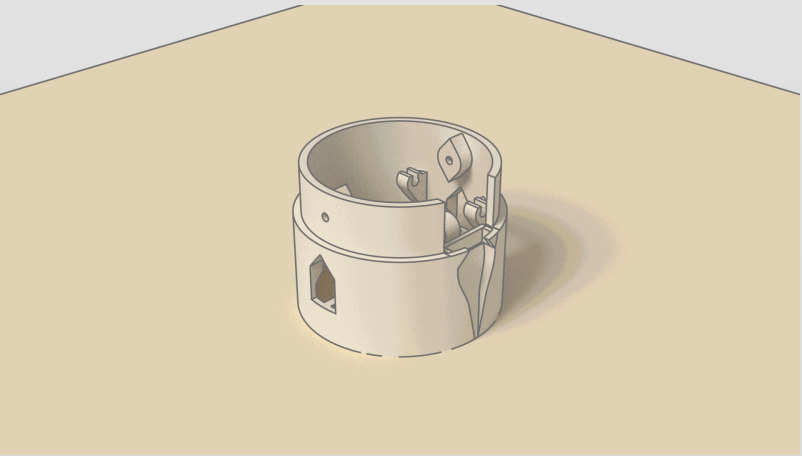
2 WALLS · 30 % INFILL · PLA AERO



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 30 % INFILL · PLA AERO

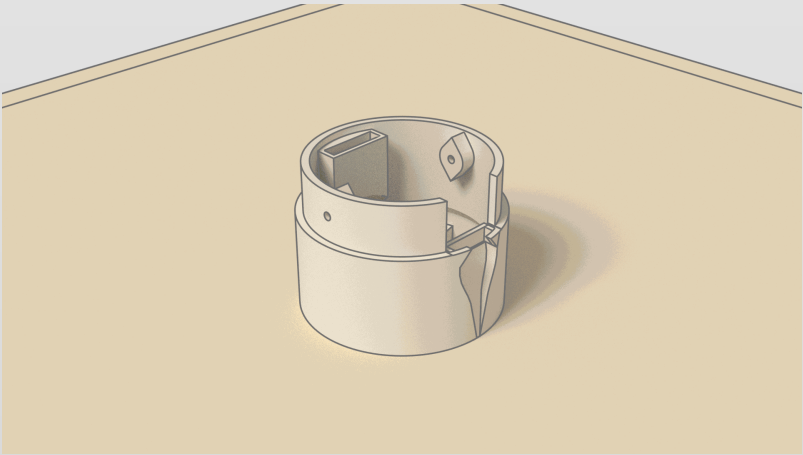


N0MD NOSE (MICRO + DOME)

FRONT FACE DOWN

2 WALLS · 30 % INFILL · PLA AERO

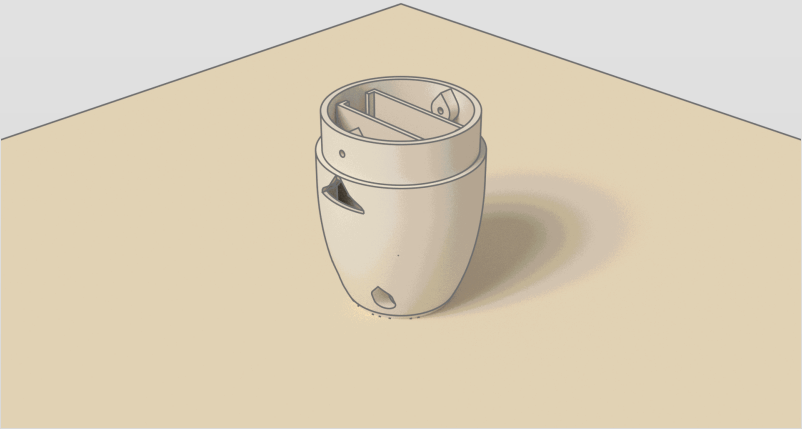
PARTS ORIENTATION



N0PD NOSE (PI + DOME)

FRONT FACE DOWN

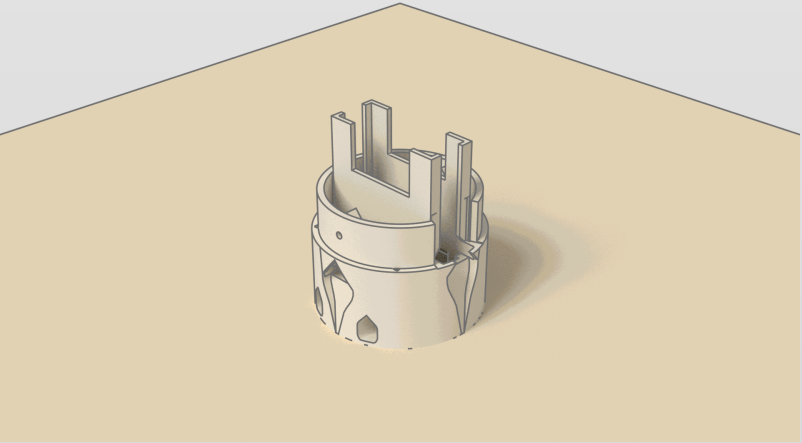
2 WALLS · 30 % INFILL · PLA AERO



N00 NOSE (DJI O4)

FACE DOWN

2 WALLS · 30 % INFILL · PLA AERO

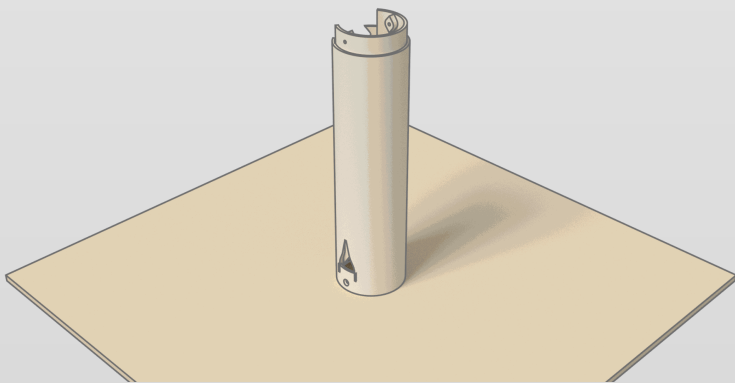


N00D NOSE (DJI O4 + DOME)

FRONT FACE DOWN

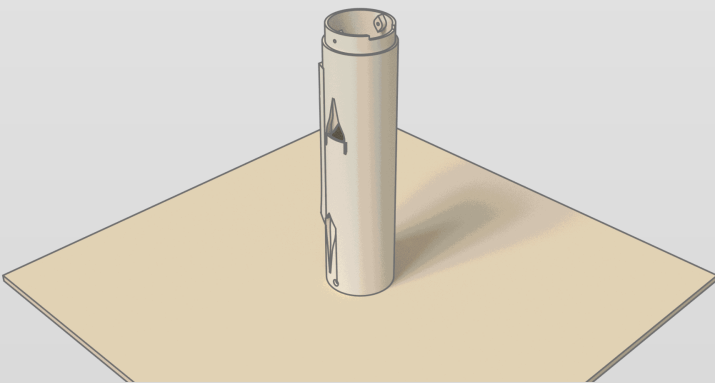
2 WALLS · 30 % INFILL · PLA AERO

PARTS ORIENTATION



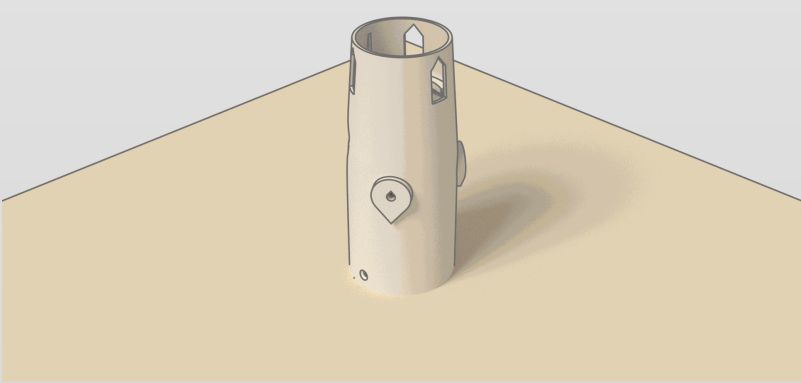
F1 FUSELAGE FRONT

FRONT END DOWN · SPIGOT UP · 5 mm BRIM
2 WALLS · 30 % INFILL · PLA AERO



F2 FUSELAGE MAIN

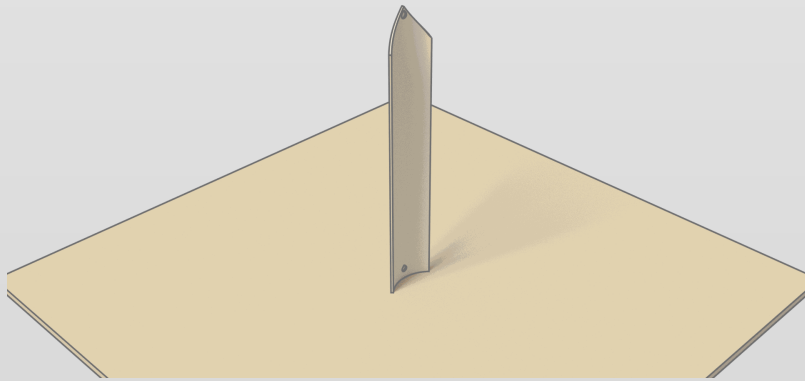
FRONT END DOWN · SPIGOT UP · 5 mm BRIM
2 WALLS · 30 % INFILL · PLA AERO



F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM
2 WALLS · 30 % INFILL · ASA AERO

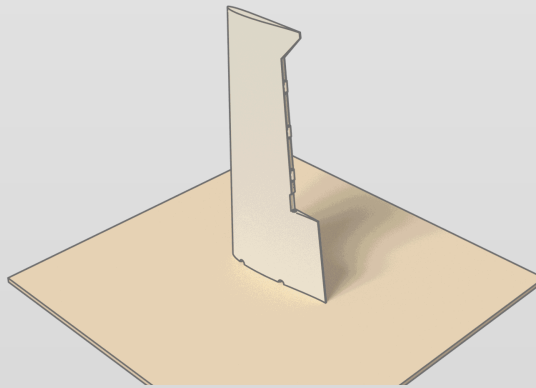
PARTS ORIENTATION



HF FRONT HATCH

STANDING, FRONT END DOWN

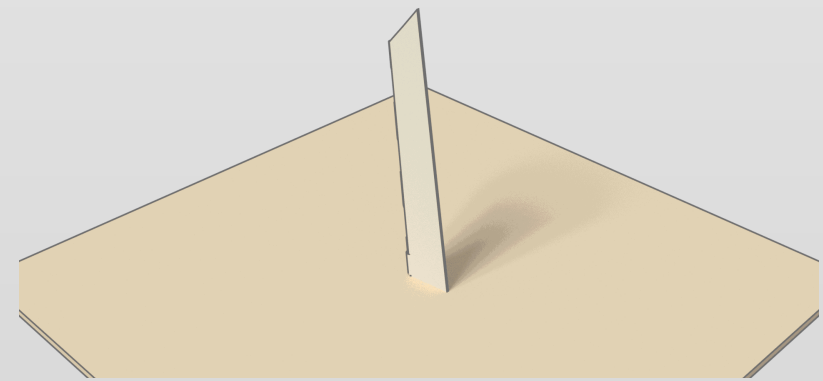
2 WALLS · 30 % INFILL · PLA AERO



WR WING HALF R/L

ON THE ROOT RIB

1 WALLS · 5 % INFILL · PLA AERO

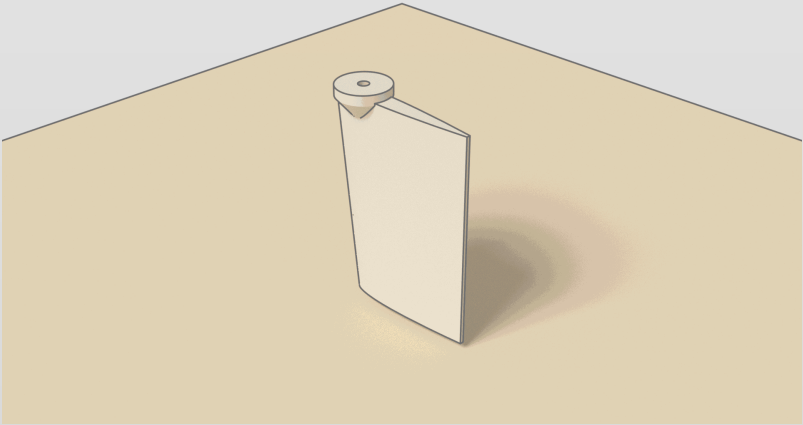


AR AILERON R/L

ON THE INBOARD END

1 WALLS · 5 % INFILL · PLA AERO

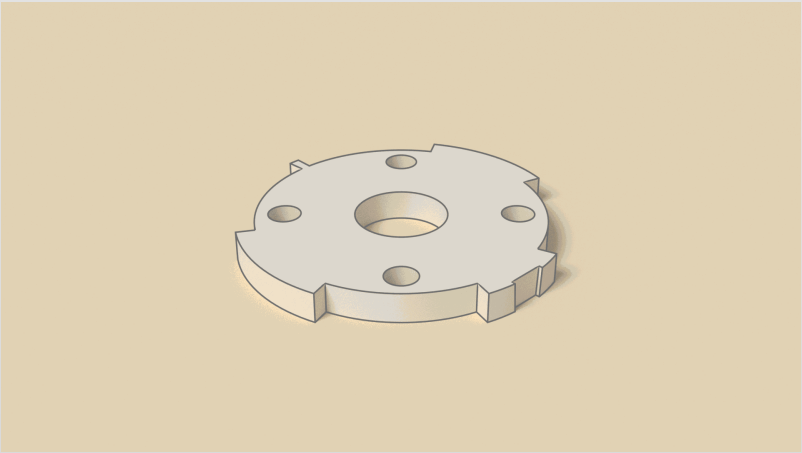
PARTS ORIENTATION



SR TAIL PANEL R / L

TIP DOWN, HUB UP

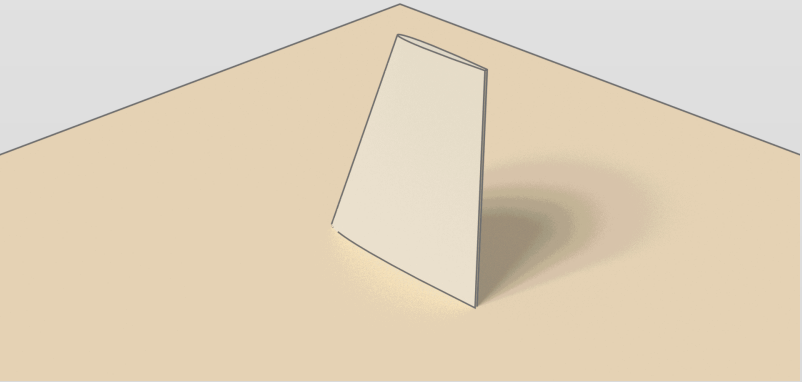
2 WALLS · 10 % INFILL · PLA AERO



M1 MOTOR PLATE

FLAT

4 WALLS · 40 % INFILL · PETG-CF

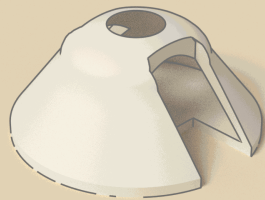


FN FIN

ON THE ROOT TAB

2 WALLS · 30 % INFILL · PLA AERO

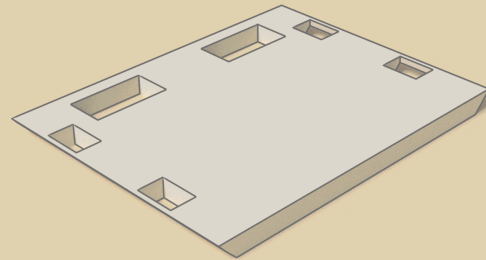
PARTS ORIENTATION



SP SPINNER

BACKPLATE DOWN

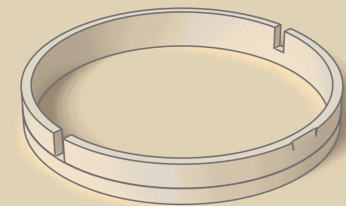
4 WALLS · 40 % INFILL · PETG-CF



VTX SHELF (NOM)

FLAT

2 WALLS · 30 % INFILL · PLA AERO

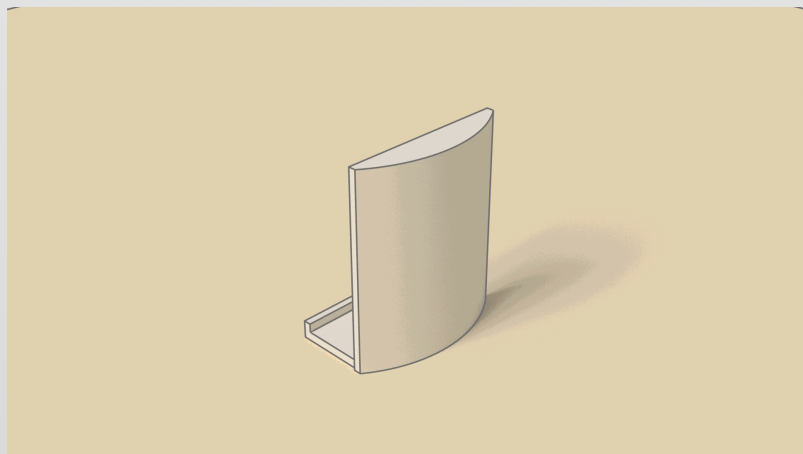


DOME RING

LIP DOWN

2 WALLS · 30 % INFILL · PLA AERO

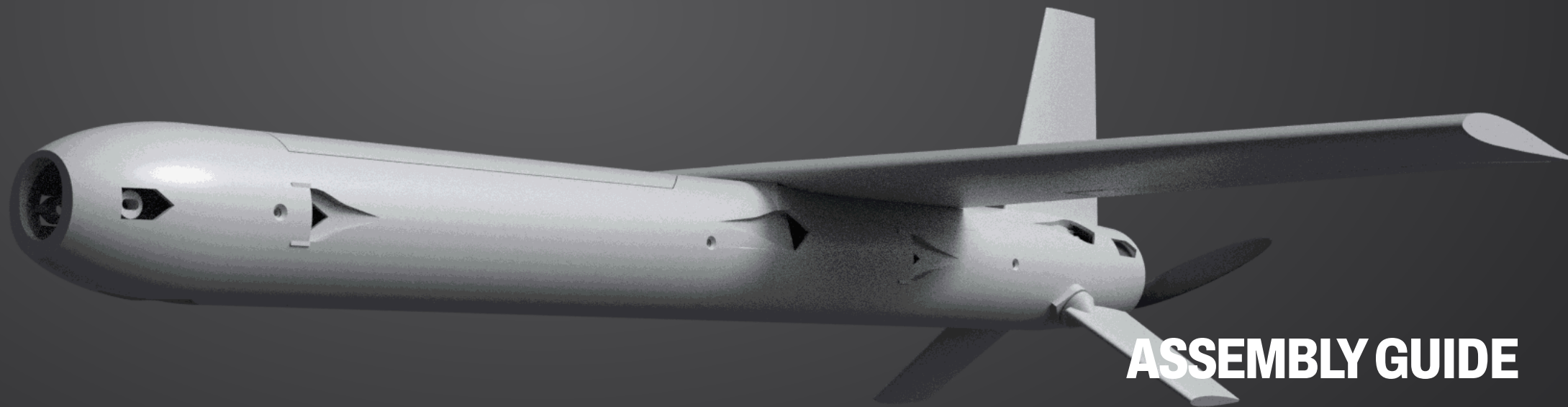
PARTS ORIENTATION



HG GPS CARRIER

ON ITS REAR WALL

2 WALLS · 30 % INFILL · PLA AERO

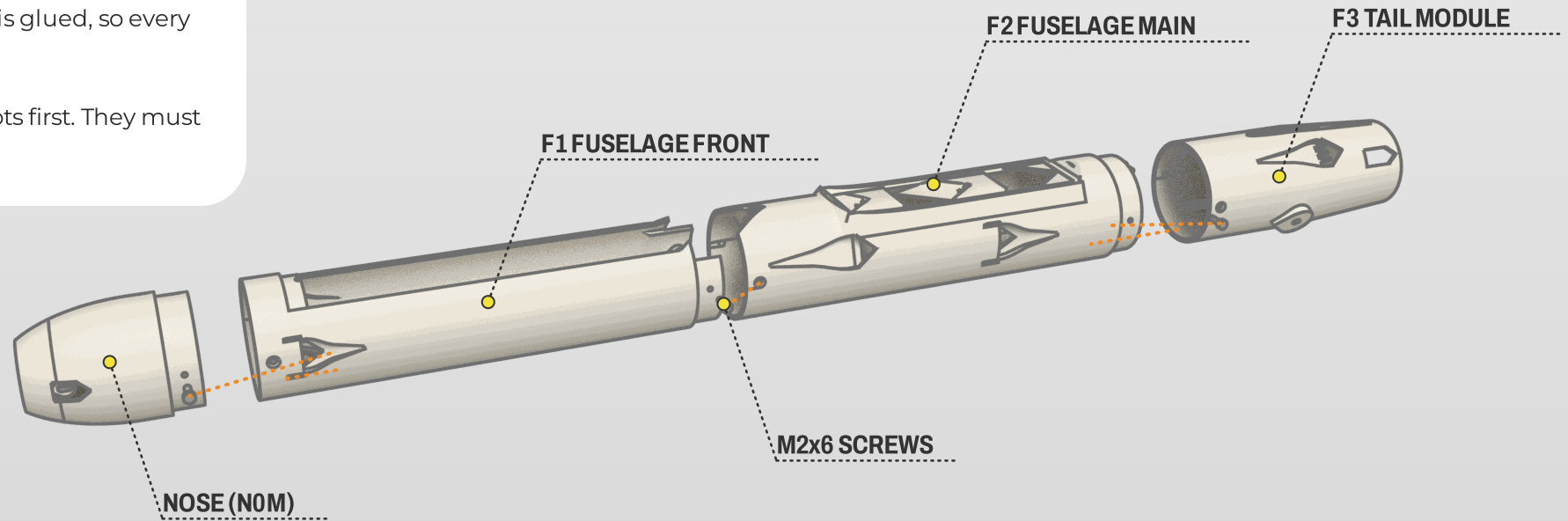


FUSELAGE ASSEMBLY



The fuselage is four modules on spigot joints: the camera nose, F1, F2 and the F3 tail module. Each joint is held by two M2 × 6 countersunk screws from the sides into bosses inside the spigot, their heads flush in the countersinks; nothing is glued, so every module comes off for service.

Clean any stringing from the spigots first. They must slide in by hand.

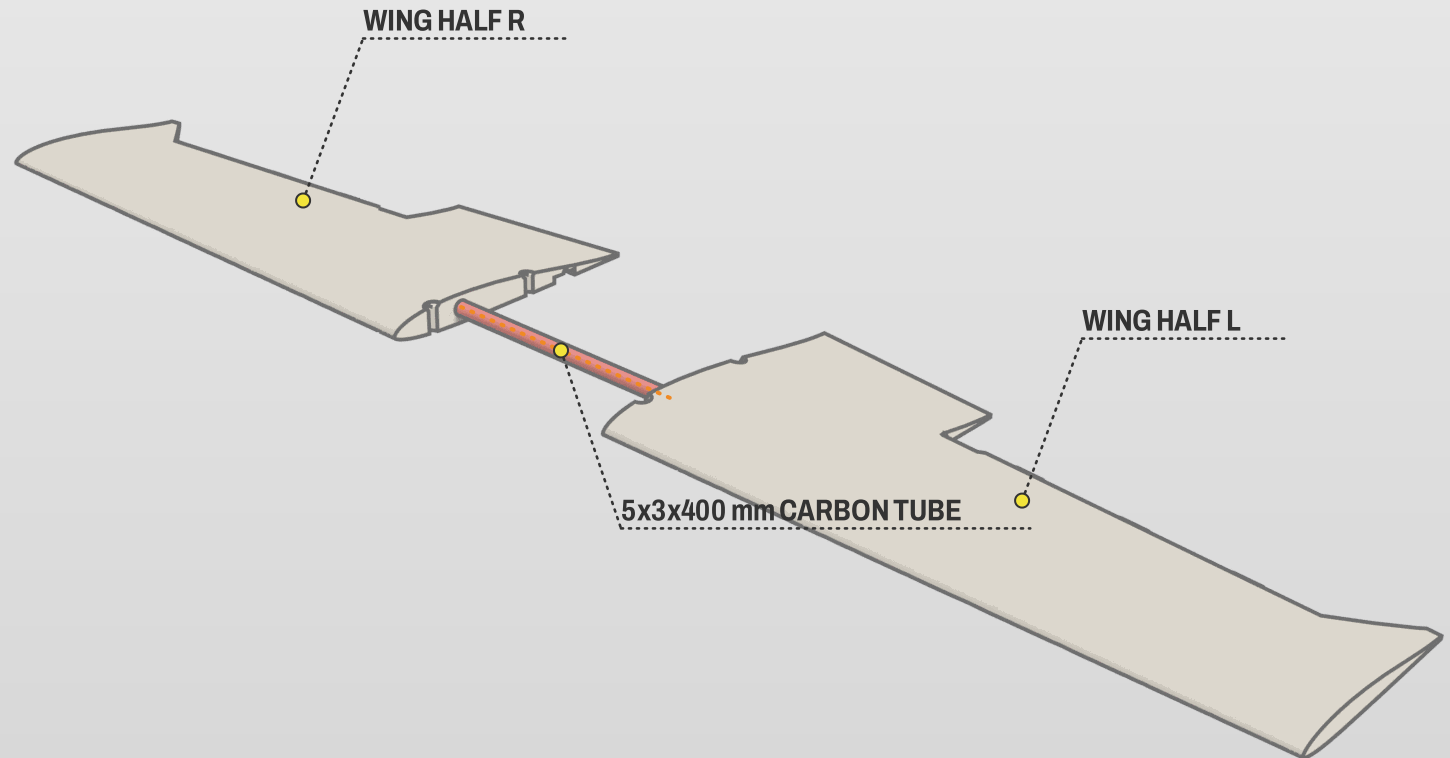


M2 × 6 SCREWS

WING ASSEMBLY



Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).

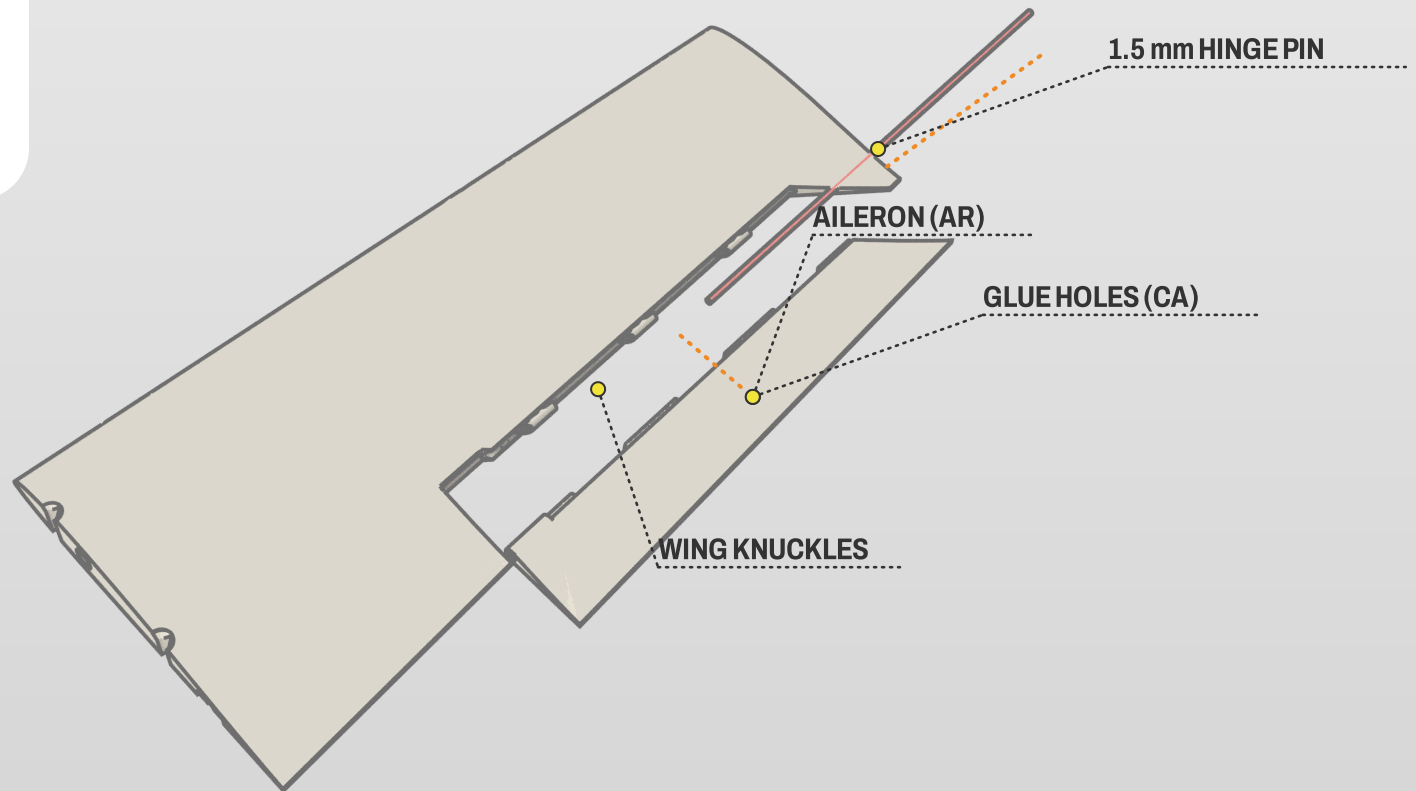


EPOXY

WING ASSEMBLY



Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.

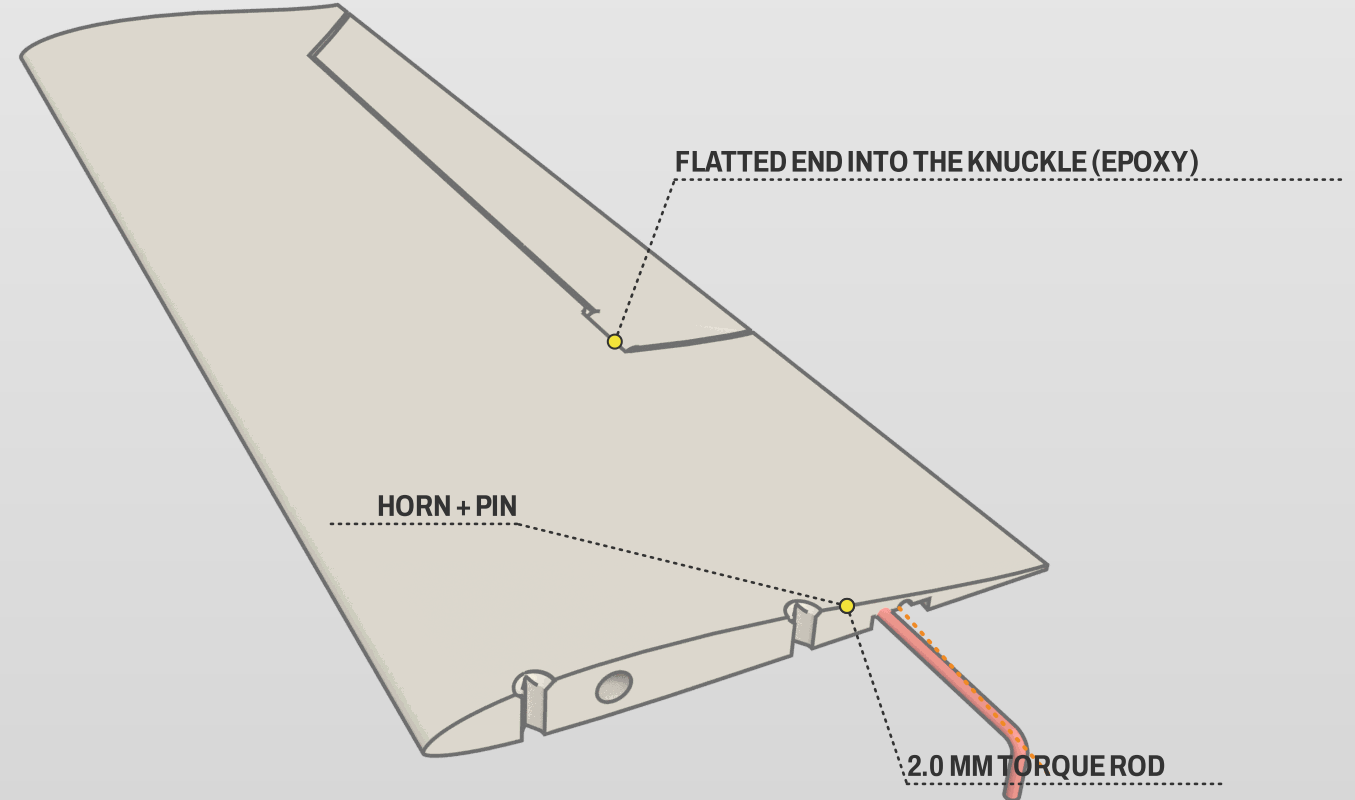


THIN CA

WING ASSEMBLY



Bend a 12 mm horn down at one end of each 2.0 mm rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.

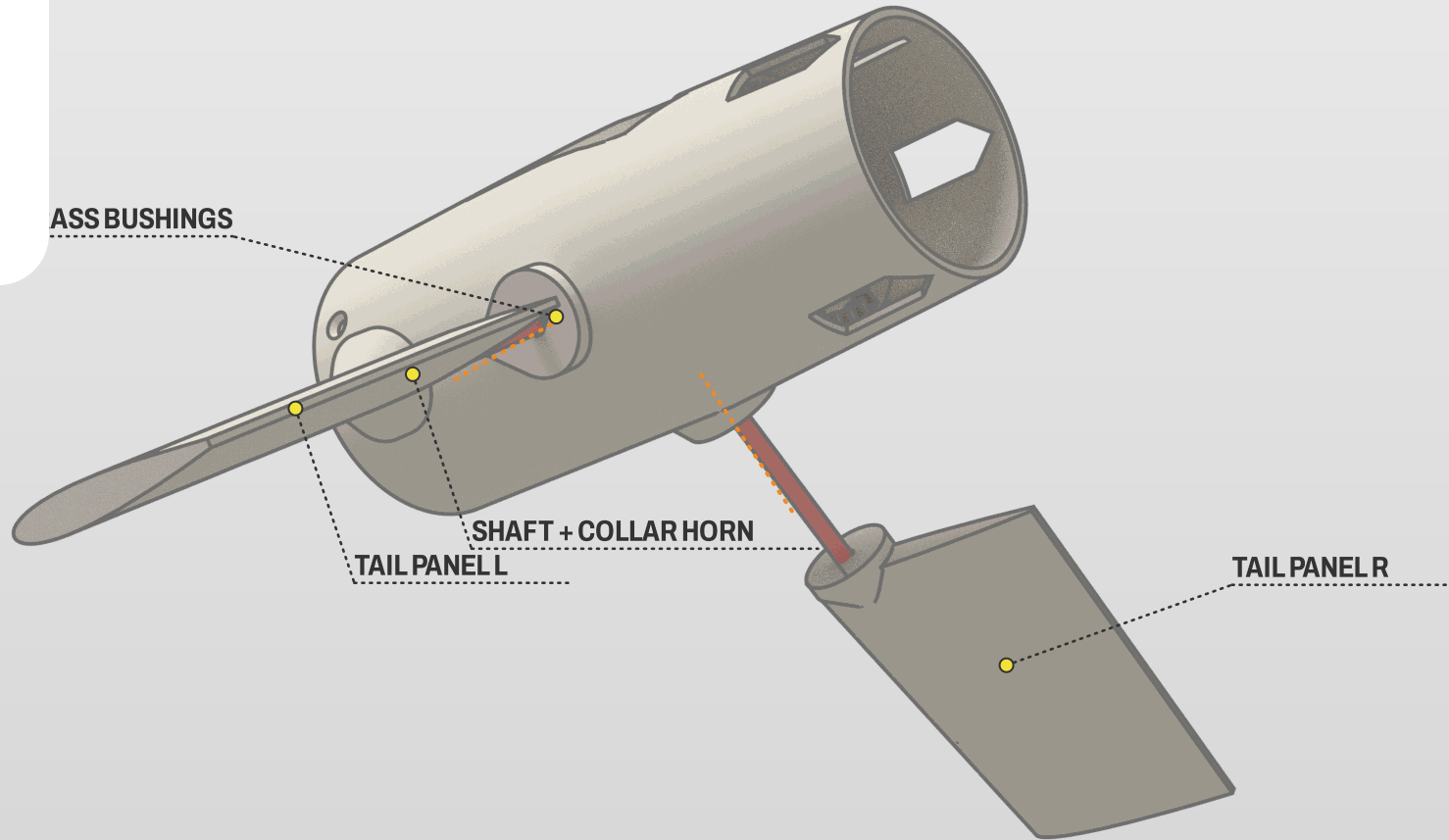


EPOXY

TAIL MODULE



Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.

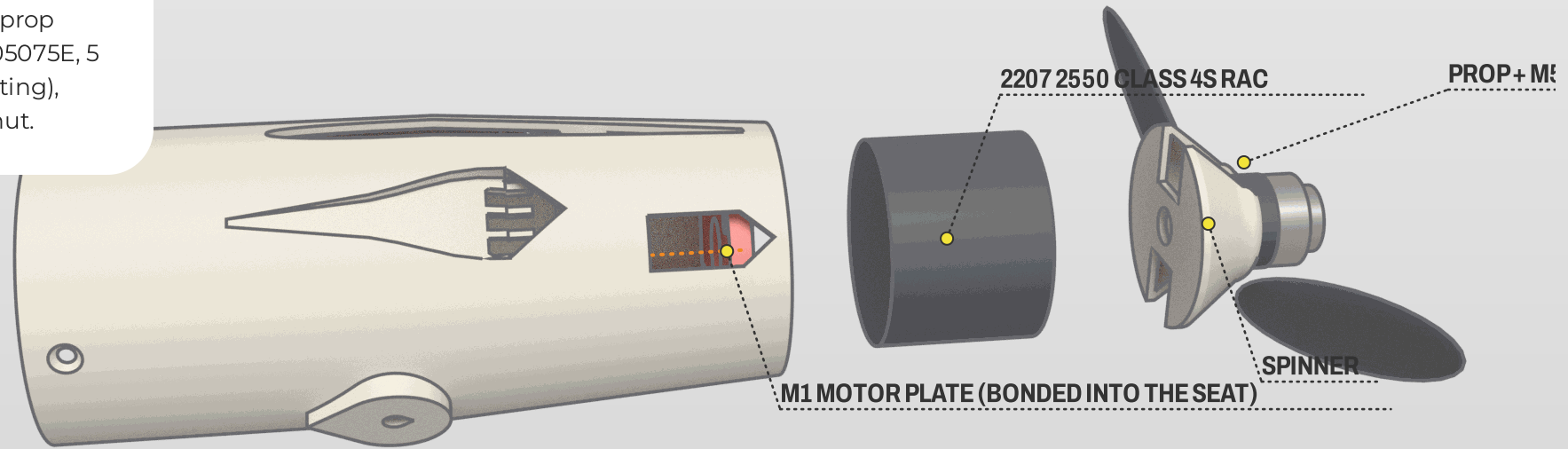


EPOXY

TAIL MODULE



Bond the M1 plate into the conical seat at the rear of F3 with high-temperature epoxy, its edge notches lined up with the cooling windows; it is the motor bulkhead. Then screw the motor to it with its 4 × M3 from inside the module, through the 28 mm opening. Do this before the tail-panel shafts and servos go in. Fit the spinner backplate, the prop (APC 5 × 7.5E fixed thin-electric pusher (LP05075E, 5 mm bore, hub 12.7 × 7.1 mm, 30,000 rpm rating), rpm-governed at 29,500) and the thin M5 nut.

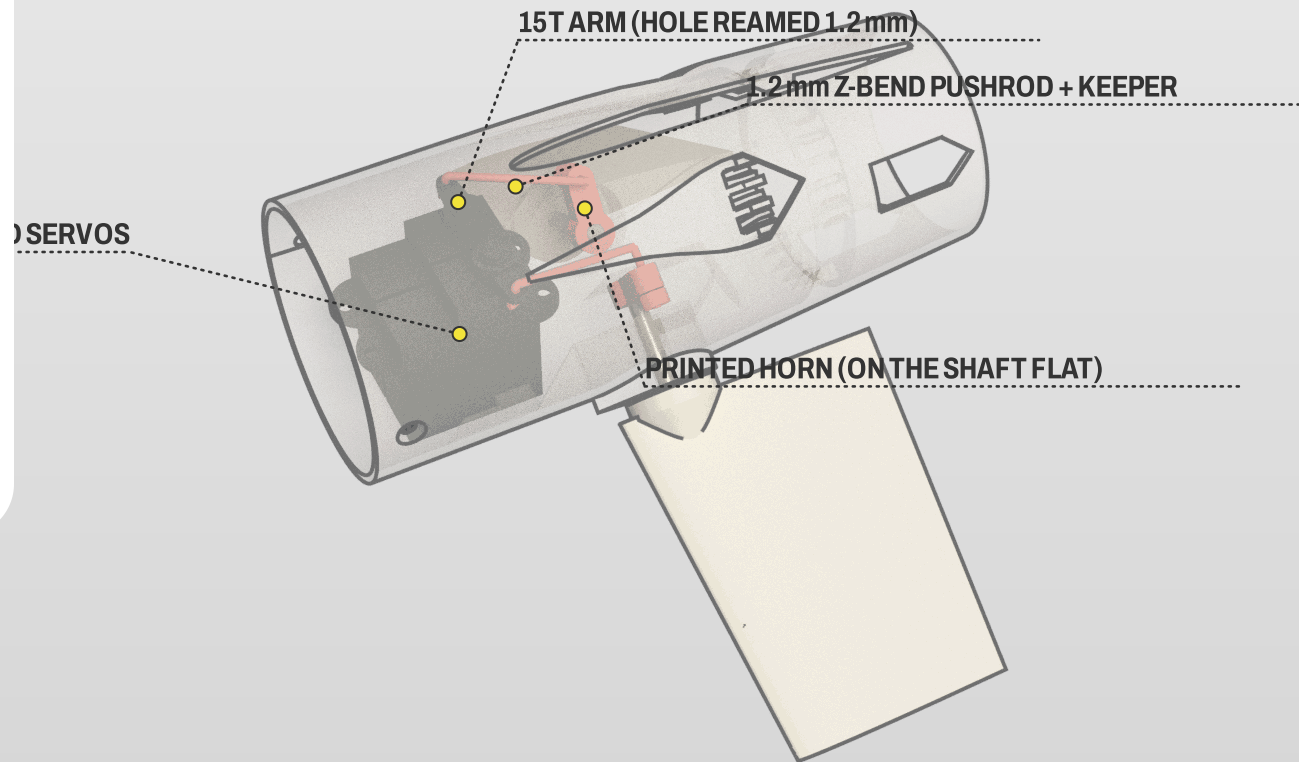


EPOXY

TAIL MODULE



Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; file a 0.3 mm flat on the inboard end of each 2.5 mm stabilator shaft and epoxy the printed horn (THR/THL) on it, arm along the panel's normal. Drill the servo arm's 6.7 mm hole 1.1 mm and ream it by hand with a 1.2 mm bit (and the horn's hole the same): the wire must turn with no play. 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend at the horn end pushed outboard through the horn's reamed hole, snap-on keeper on the protruding end (unclip it to take the tail module off). Free play at each panel trailing edge ≤ 0.2 mm. Check both panels move freely through $\pm 20^\circ$.

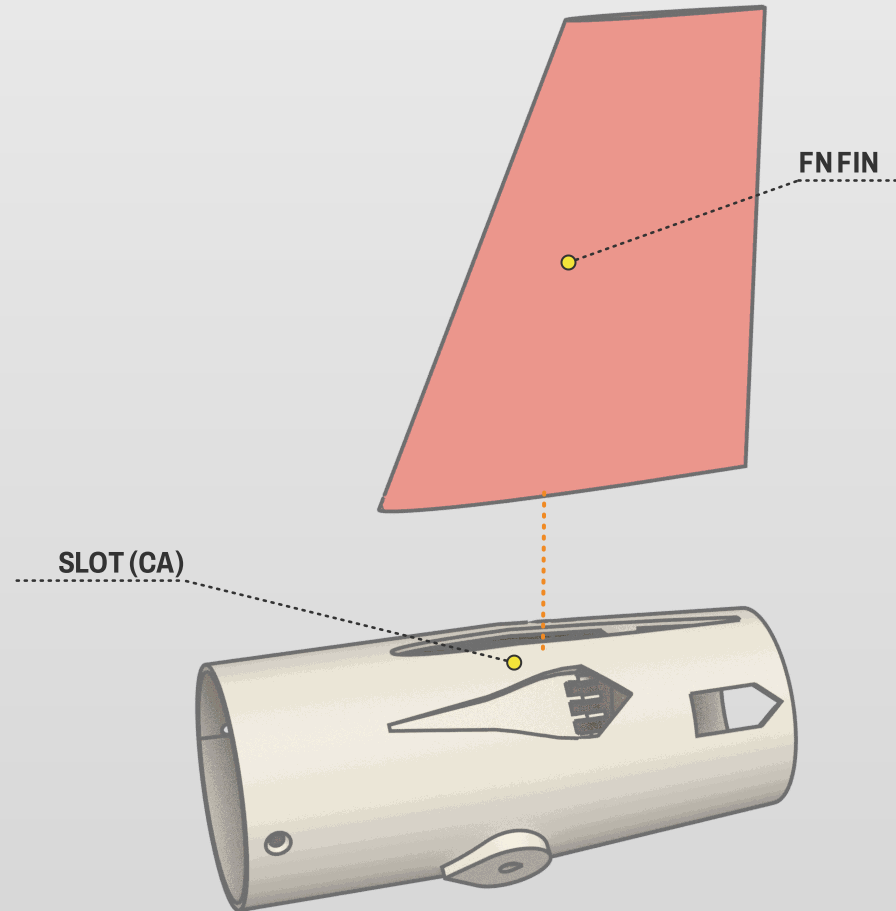


SERVO TAPE

TAIL MODULE



Glue the fin into the top slot of F3 with CA, vertical and square to the wing.

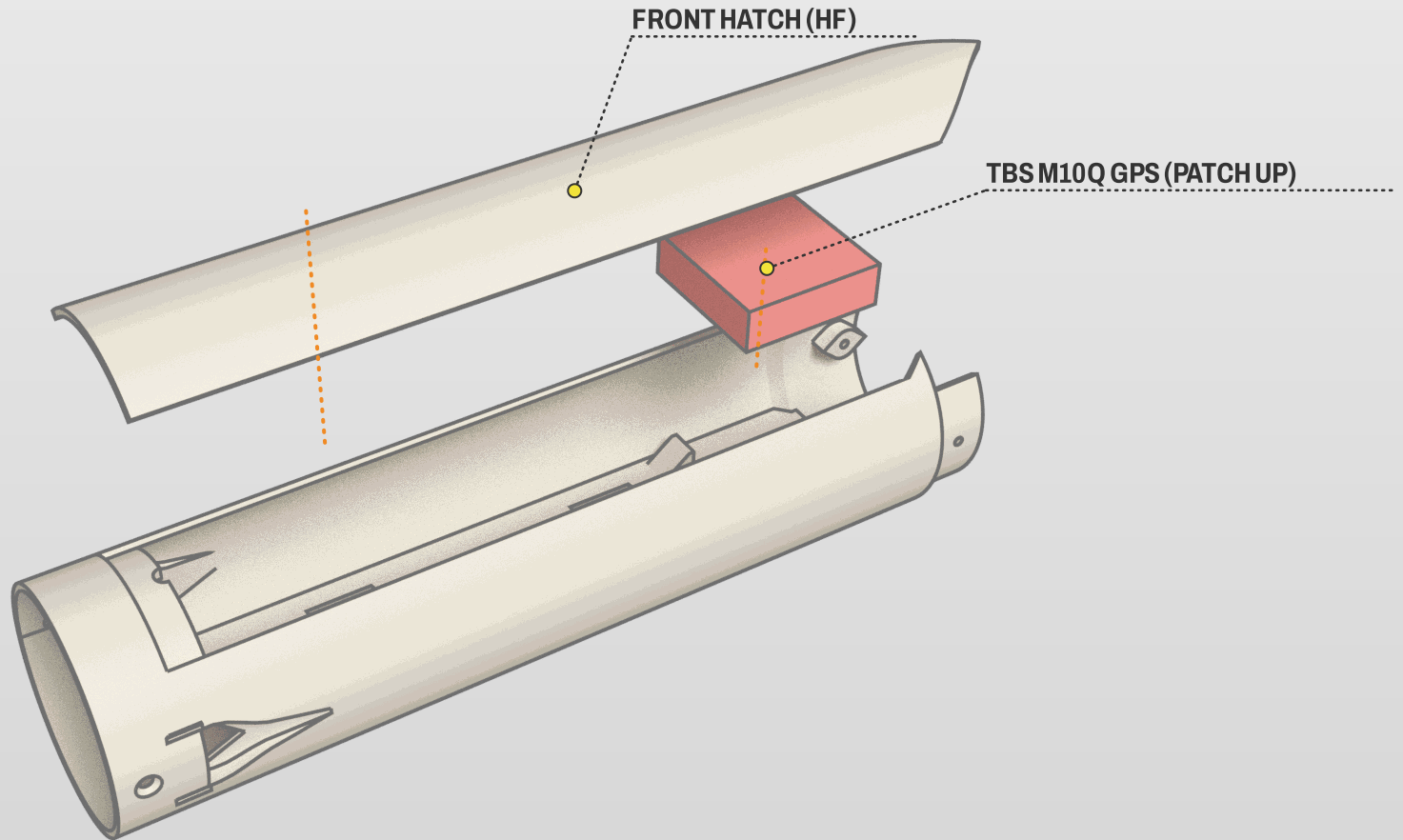


CA GLUE

FUSELAGE ASSEMBLY



Bond the printed GPS carrier (HG) under the rear of the front hatch skin. Foam-tape the TBS M10Q GPS, patch up, to its ceiling, against its rear ledge. The hatch's pointed rear end reaches into F2 and its rear screw goes into F2: take the hatch off before separating F1 and F2. Press the magnets into the hatch lips and the hatch, checking polarity. Leave a slack loop in the GPS lead so the hatch lifts off.

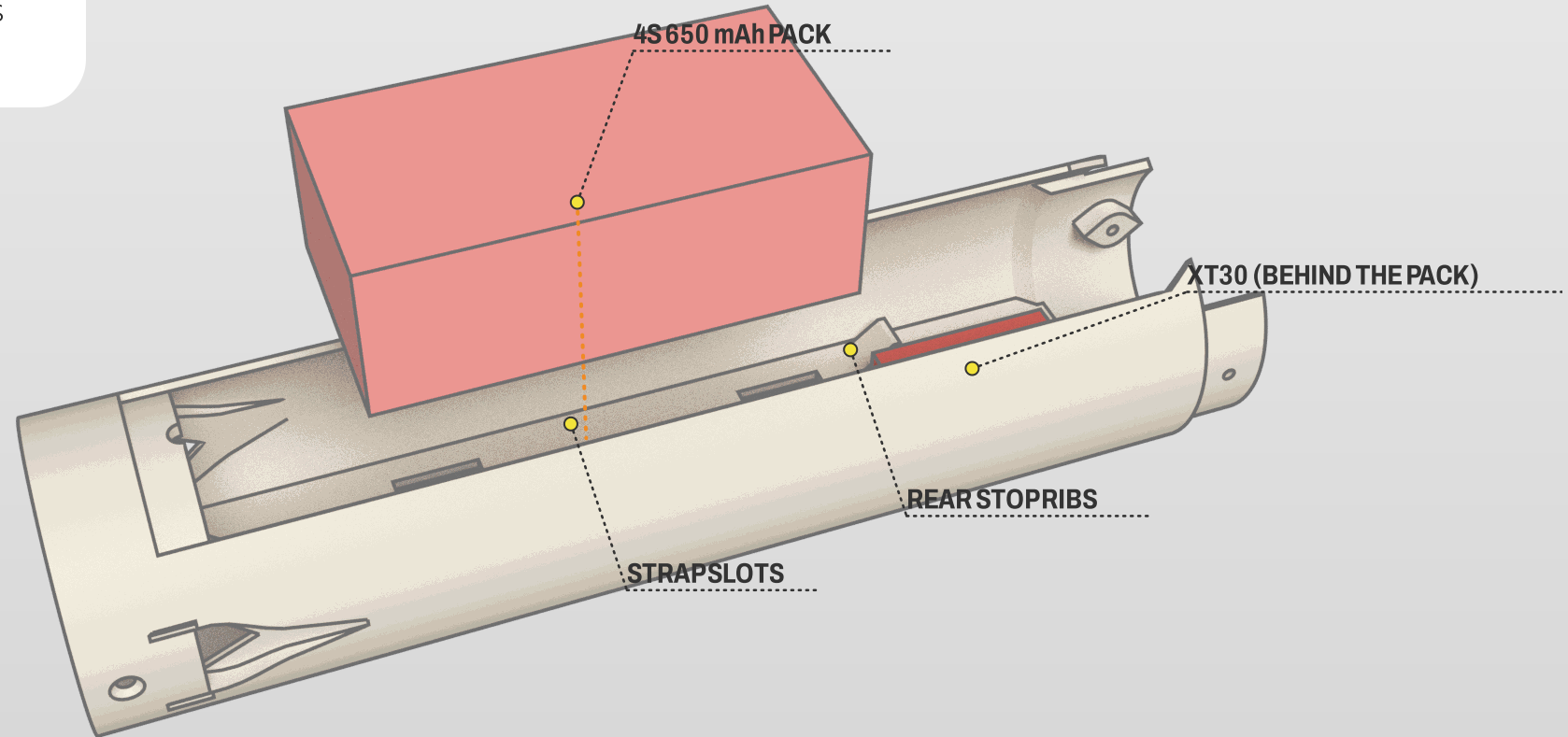


FOAM TAPE

FUSELAGE ASSEMBLY



The pack sits on the FI floor with a strap through the slots, leads at its rear. Two floor ribs are its rear stop; the XT30 pair lies behind the pack under the GPS and unplugs through the front hatch.

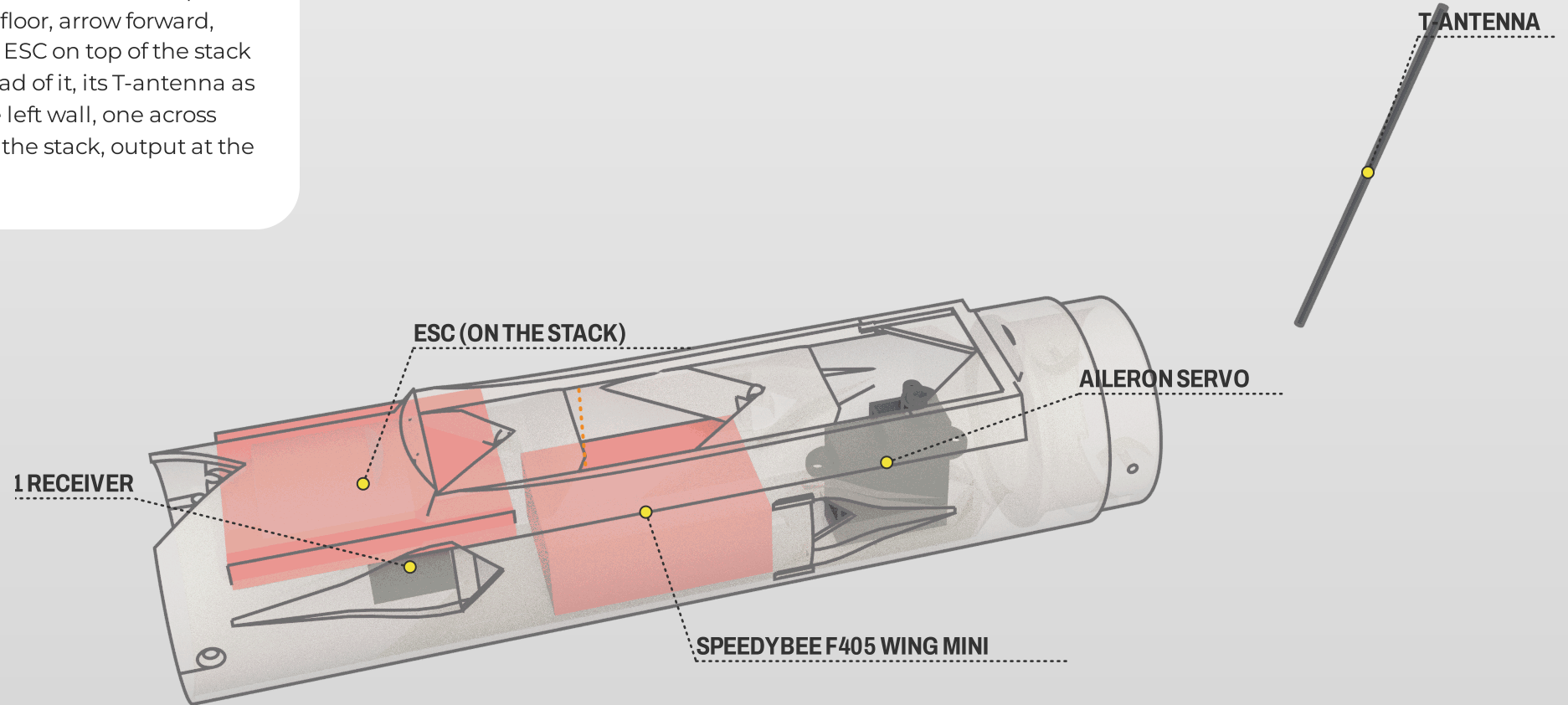


VELCRO STRAP

EQUIPMENT



SpeedyBee F405 WING MINI stack on foam tape and a strap on the wing-bay floor, arrow forward, servo BEC jumper at 6 V; the ESC on top of the stack on a foam pad. Receiver ahead of it, its T-antenna as an L at $x \approx 220$ (one leg up the left wall, one across the floor). Aileron servo aft of the stack, output at the aft end.

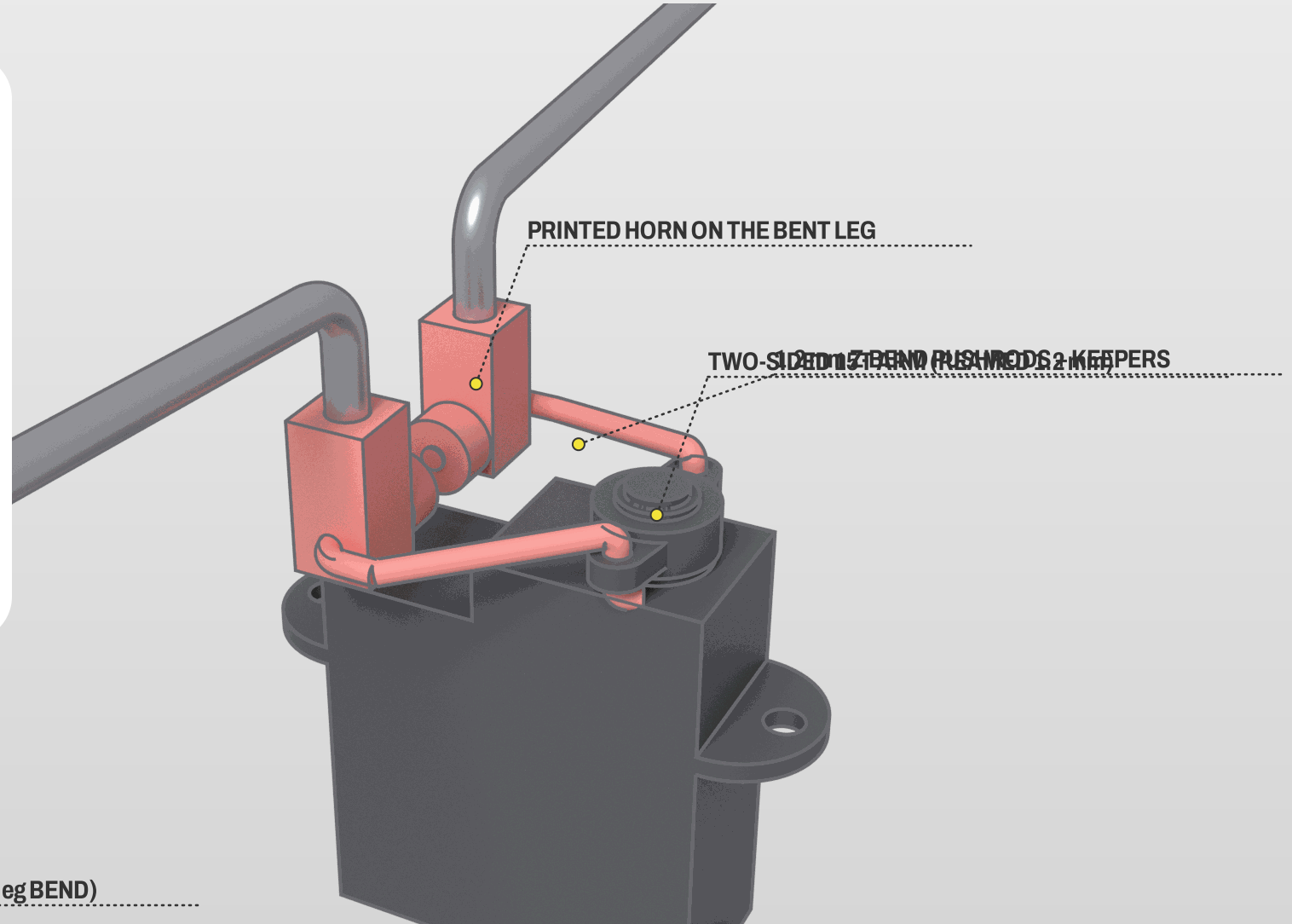


FOAM TAPE

EQUIPMENT



ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque rod has one 90° bend (the horn leg, 6 mm from the centreline); epoxy the printed horn (AHR/AHL) on the leg, 4 mm deep. Drill the arm's 4.3 mm holes 1.1 mm and ream them by hand with a 1.2 mm bit (and both horns' holes): the wire must turn with no play. Each horn connects to the arm with a 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend pushed inboard through the horn's reamed hole, snap-on keeper on the protruding end. Check: with the servo powered, each aileron trailing edge moves ≤ 0.2 mm (free play lowers aileron flutter to about 33 m/s). Set both ailerons 3° trailing edge up at servo neutral. No play at either end.

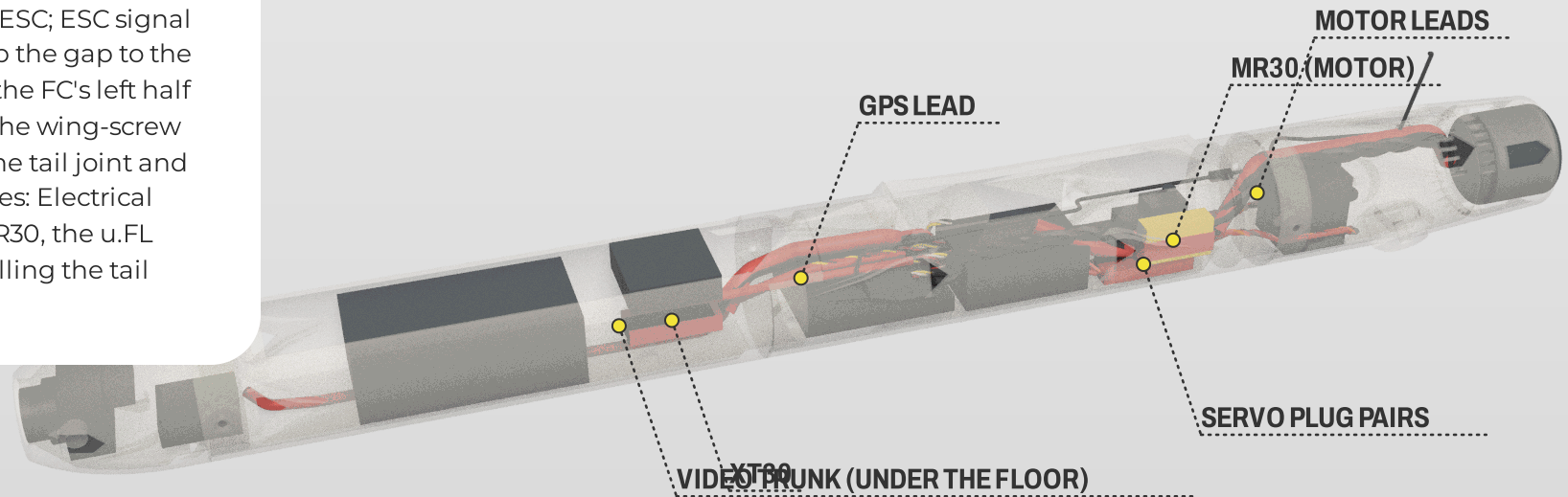


Z-BENDS

EQUIPMENT



ESC flat on its foam tape at the bottom of the wing bay, receiver on top of it, FC stack 5 mm behind it (servo header row on the right). Battery lead (14 AWG) from the XT30 over the ESC and down the gap to the FC power board's BAT pads; 14 AWG ESC feed from its ESC pads up onto the ESC; ESC signal (M1, GND, TLM) and receiver lead up the gap to the FC; motor phases up the gap, over the FC's left half to the MR30; the ELRS coax under the wing-screw bridges to the u.FL joint ahead of the tail joint and up into the fin. Cut lengths and lanes: Electrical Integration manual. Unplug the MR30, the u.FL joint and the servo plugs before pulling the tail module.

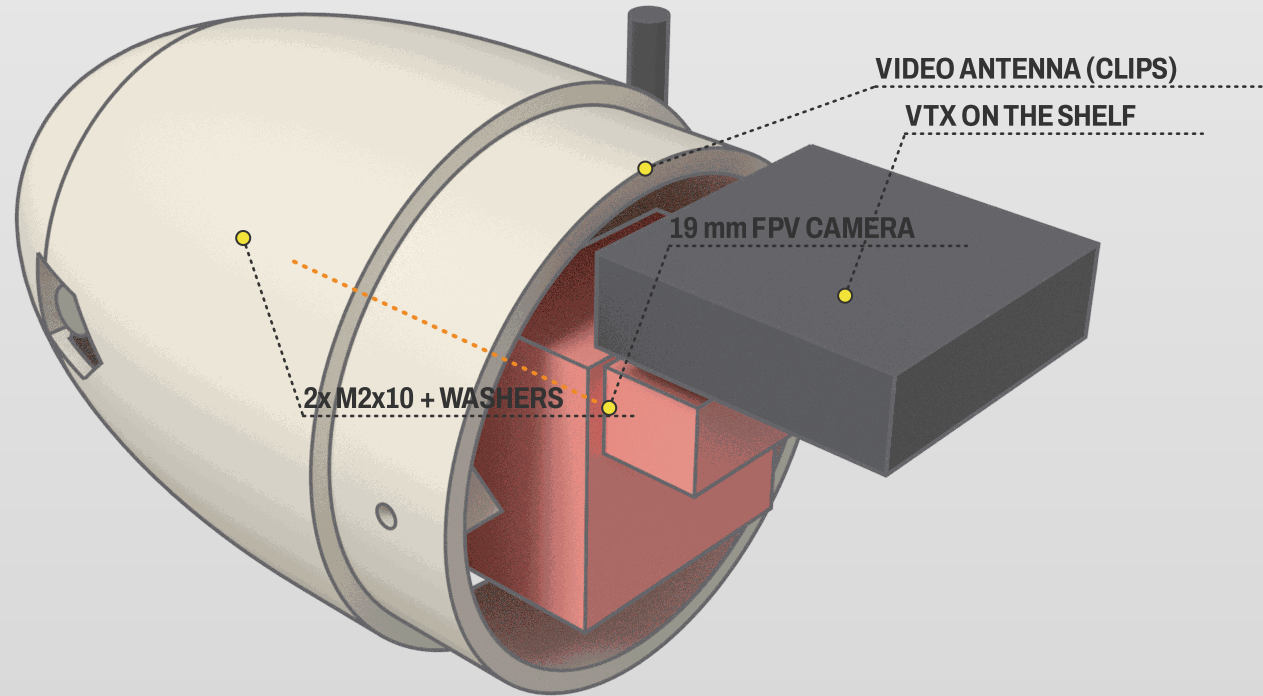


HEAT SHRINK

CAMERA NOSE



Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).

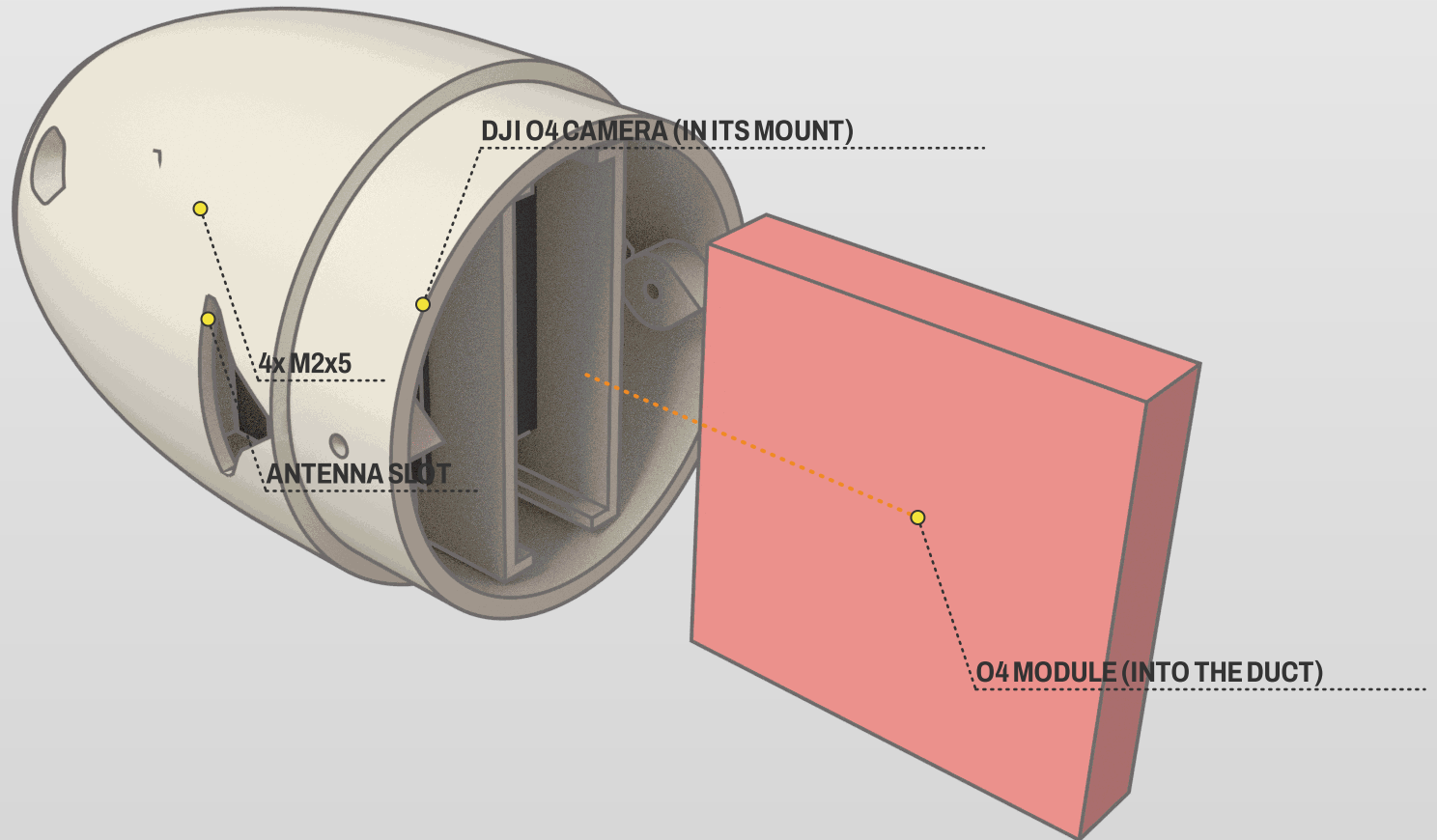


FOAM TAPE

CAMERA NOSE



DJI O4 option: clamp the camera in its mount with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount). Plug the coax and 3-in-1 lead into the module and slide it edge-first into the duct until it stops; foam strip in the top groove, tape over the rear edge, never over its faces.

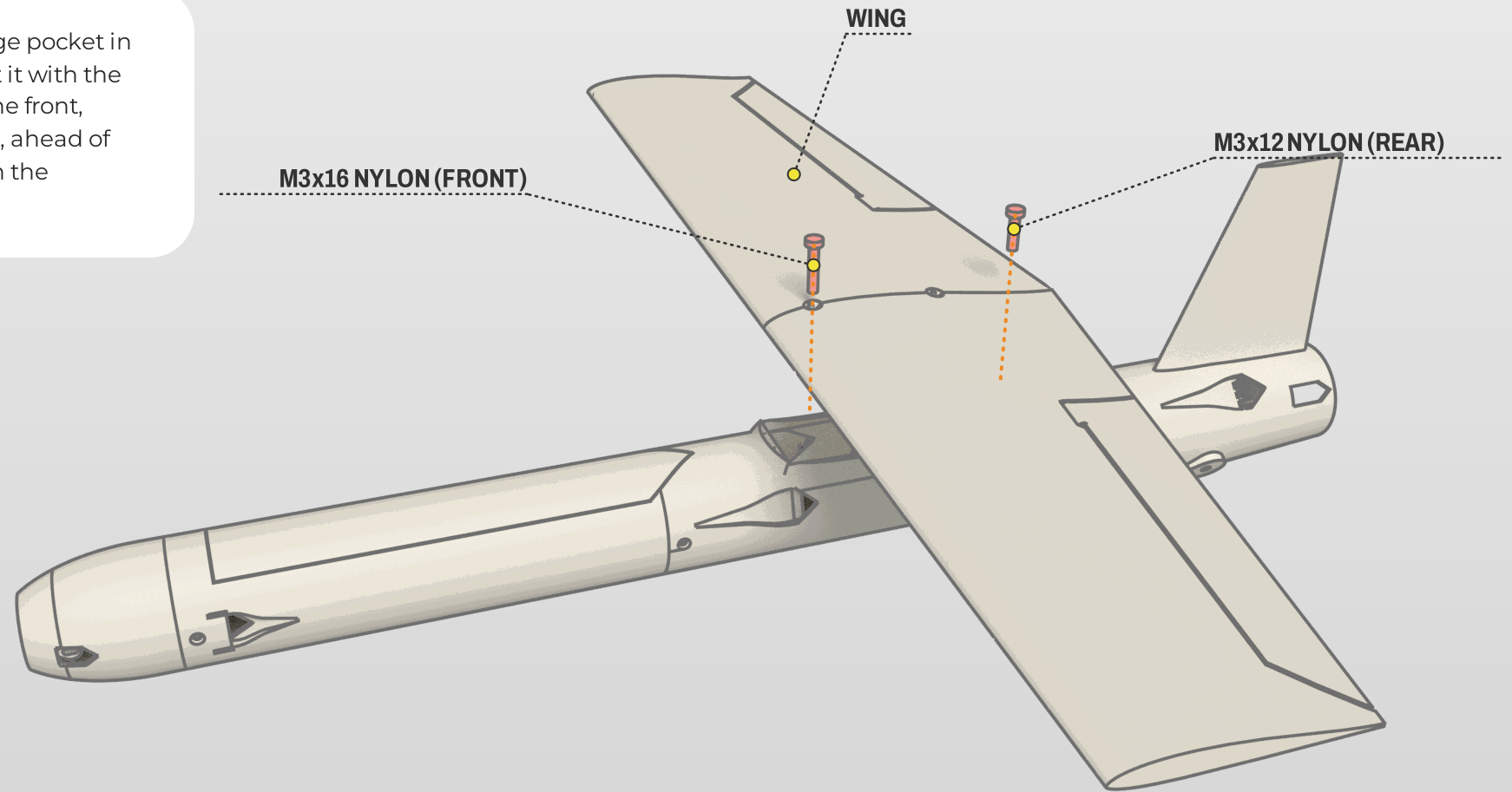


M2 × 5 SCREWS

FINAL ASSEMBLY



Push an M3 nut up into each bolt-bridge pocket in F2. Set the wing on the saddle and bolt it with the nylon screws: Countersunk M3 × 16 at the front, countersunk M3 × 12 at the rear (LE + 58, ahead of the aileron horns); the heads sit flush in the countersinks.



NYLON SCREWS

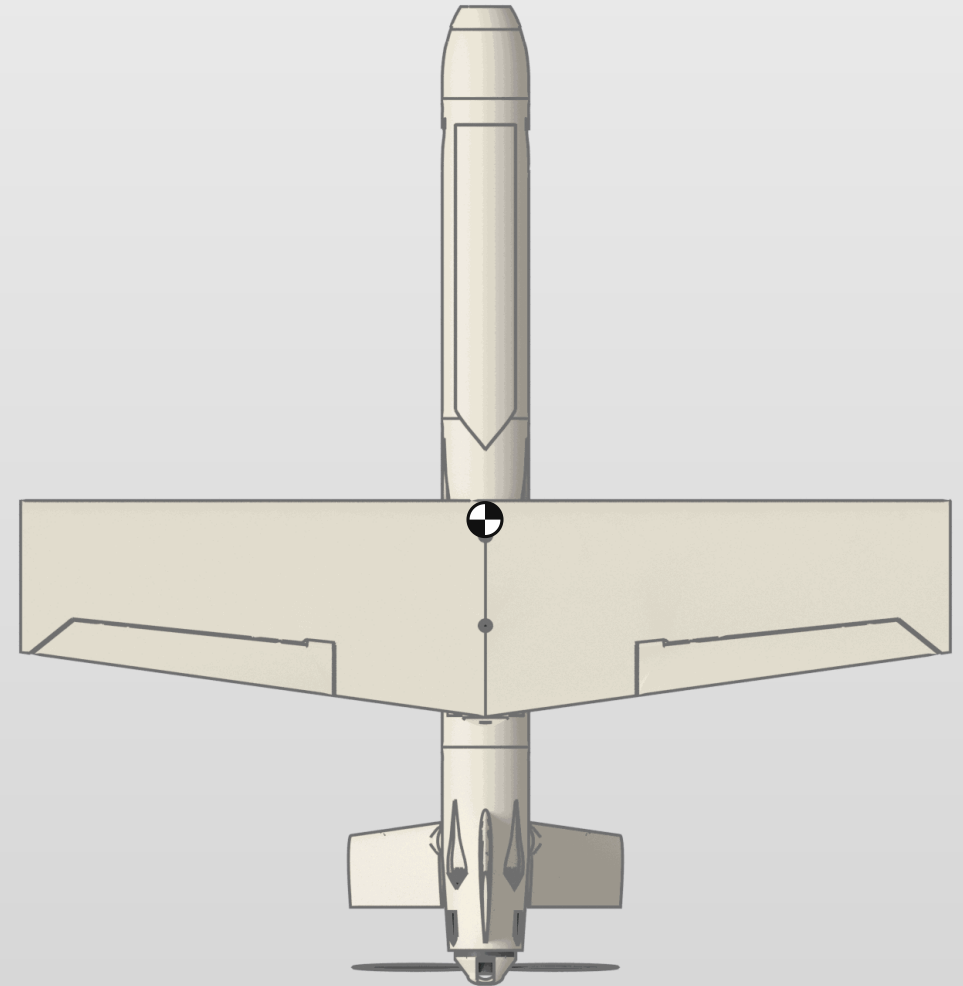
FINISHING BUILD



Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws $\pm 10^\circ$ on the tail panels, about $\pm 10^\circ$ (± 3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

Balance: CG 9 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose or tail ballast cup.

Glass skin: one $\pm 45^\circ$ layer of 49 g/m^2 glass on the wing halves and tail panels after printing, then a thin paint finish.

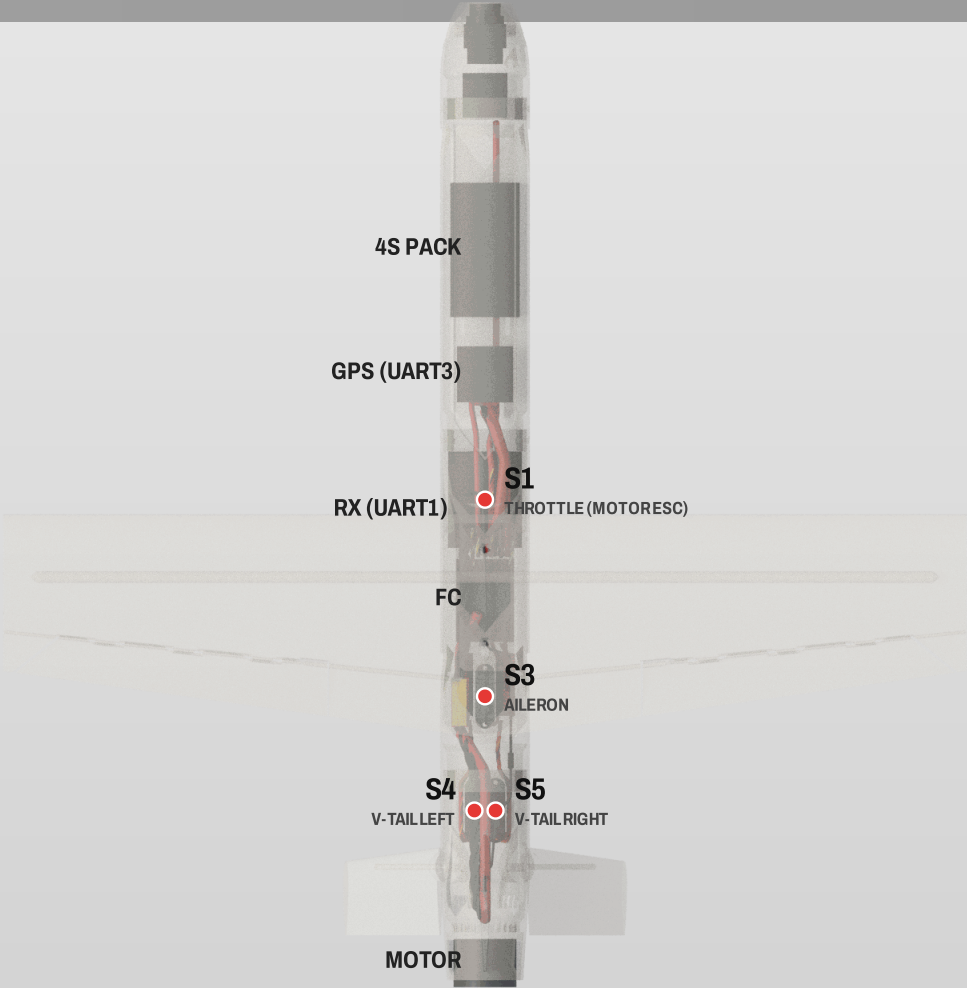


WIRING DIAGRAM



The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	ESC (DShot300, bidirectional)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)



ELECTRICAL INTEGRATION

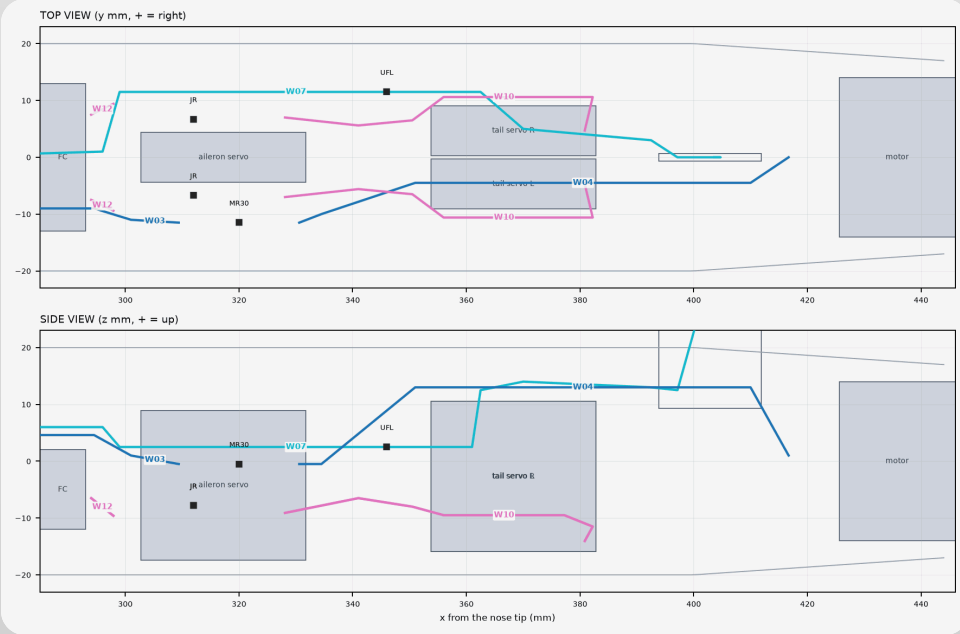
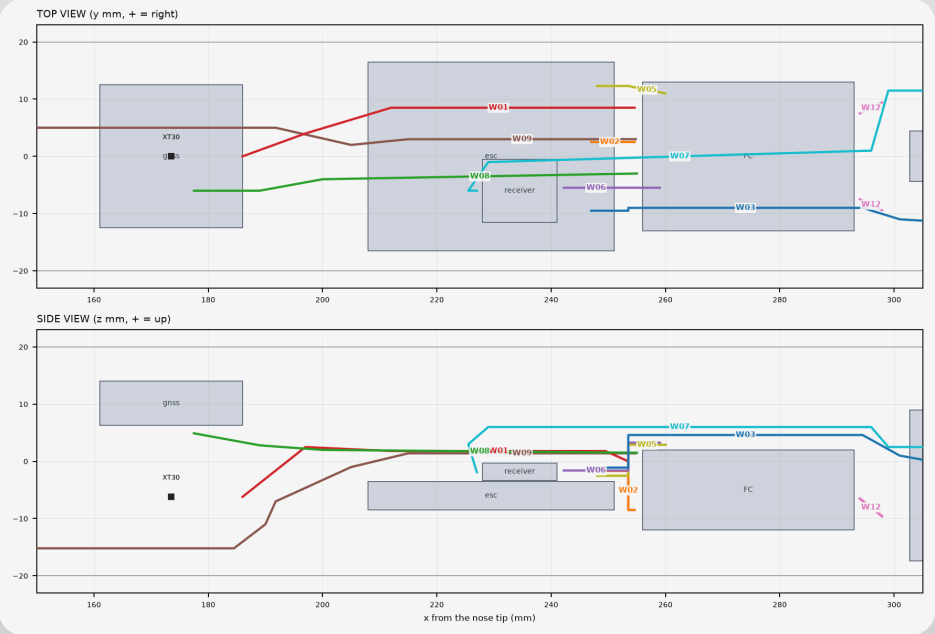
Battery → FC power board (80 A current sensor) → ESC (motor on M1) → MR30 at the tail joint → motor. Main DC path 14 AWG (46 A on a pass, 51 A static), motor phases 16 AWG (38 A RMS). Cable lengths are measured on the 3D harness model; the cut list adds the termination allowances and slack. Full detail, pinouts, settings and tests: the **Electrical Integration manual** (PDF in the package and on the web page).

ID	CABLE	FROM	TO	AWG	CUT mm
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under...	FC power board BAT+ / BAT- pads	14	110
W02	ESC feed	FC power board ESC+ / ESC- pads	HDZero Halo 4in1 70A ESC AM32 BAT+ / BAT- pads (XT60...	14	35
W03	Motor phases, ESC side	HDZero Halo 4in1 70A ESC AM32 M1 pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the...	16	95
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat...	16	140
W05	ESC signal	HDZero Halo 4in1 70A ESC AM32 8-pin SH1.0 lead (only M1,...	FC S1 header (signal, GND); TLM to the R4 pad	28	45
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the ESC	FC receiver header (5V, GND, RX1, TX1)	28	45
W07	ELRS antenna coax	RP1 V2 u.FL	Dipole feed at the centre of the fin dipole (z 39.5)	coax	265
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC GPS interface with compass (4V5, GND, RX3, TX3, SDA,...	28	115
W09	Video trunk (NOM nose)	Nose joint plug (camera on the nose; VTX on the nose or,...	FC pads: Vv, G, C1 (camera in), VTX (video out)	28	265
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	26	stock
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	26	stock
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	26	stock

ELECTRICAL: POWER AND ROUTING

CABLE	CIRCUIT	CURRENT A (PASS / STATIC)	WIRE	DROP V	dT K / PASS	CONNECTOR RATING
W01+W02	Battery to ESC (DC)	46 / 51	14 AWG	0.07 (0.5 %)	20	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	38 / 42	16 AWG	0.17 (1.1 %)	33	MR30: 30 A cont / 40 A for 1 min per contact

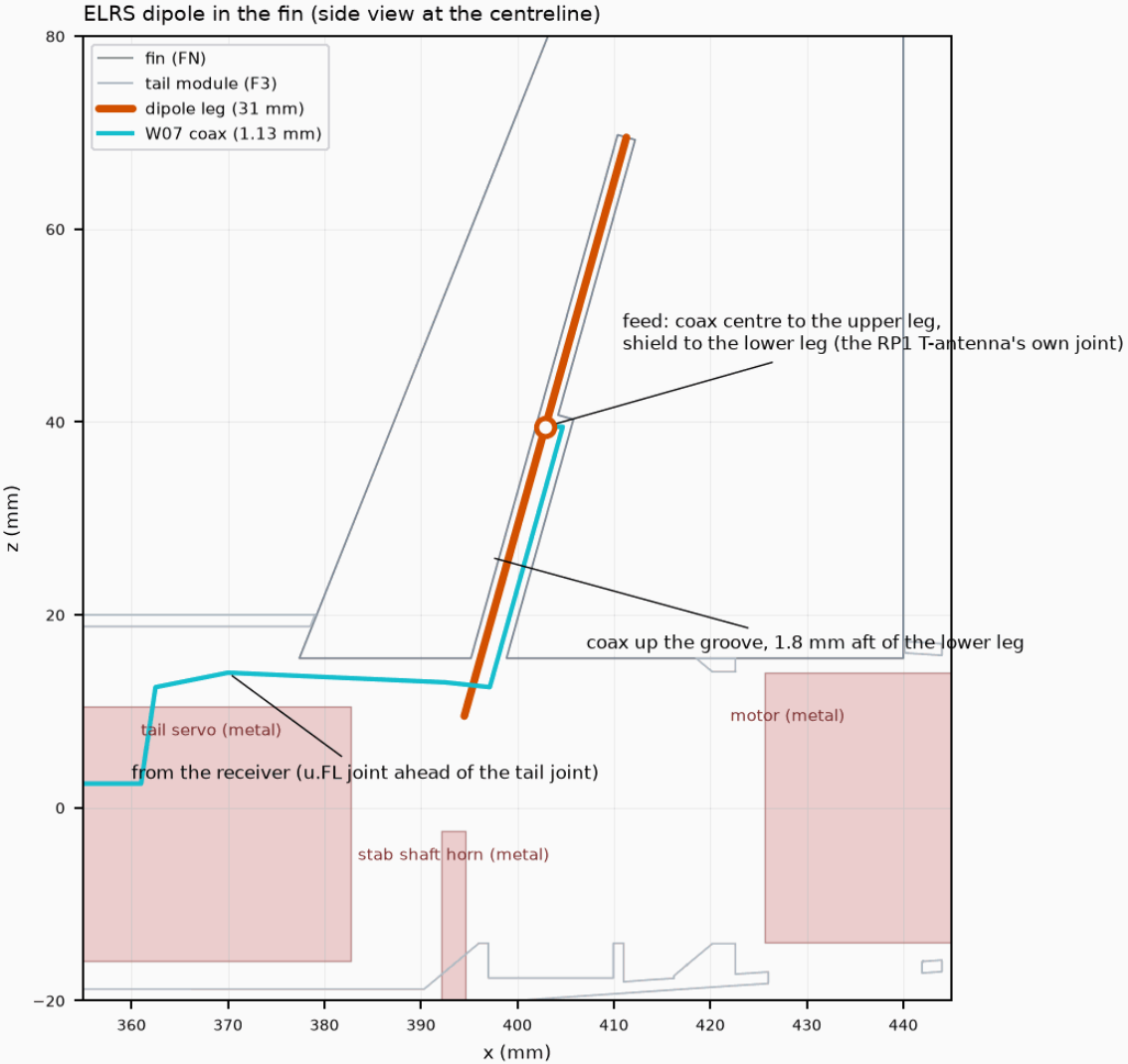
XT30: rated 15 A continuous / 30 A peak. Flown as the test base (70 % throttle cap, no governor) the pack draws about 18 A: fine. Only with the governor fitted (46–51 A) does it need an XT60 lead and a ready-made XT60 pack. Never connect the ESC lead's VBAT pin to the FC's servo header or a 5 V pad. Power-board jumpers: servo BEC 6 V; Vv battery voltage only if the camera and VTX accept 4S.



ELECTRICAL: RECEIVER ANTENNA



The receiver's antenna is a vertical dipole inside the fin, clear of all metal: the RP1 V2 T-antenna's two 31 mm legs lie in a groove on the fin's 30 % chord line with the T's joint (the feed) at the centre. The antenna's own coax runs down a parallel groove beside the lower leg into the tail module, to an in-line u.FL joint just ahead of the tail joint; a 1.13 mm u.FL extension continues to the receiver on the ESC. Unplug the u.FL joint with the MR30 and the servo plugs before pulling the tail module. Lay antenna and coax in the grooves before the fin is glued; three dots of epoxy, tape over the grooves.



ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 90 on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

Firmware: ArduPlane 4.5 or later · **Board:** SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING)

The setup document and the starting .param file are in the package (ArduPilot folder). Verify every value on the bench before flight.



FLIGHT TEST AND SPEED STAGES



The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	27 %	24 m/s
35 m/s	32 %	29 m/s
40 m/s	36 %	33 m/s
45 m/s	41 %	38 m/s
50 m/s	45 %	41 m/s
55 m/s	50 %	46 m/s
60 m/s	54 %	49 m/s
65 m/s	59 %	54 m/s
70 m/s	64 %	58 m/s
75 m/s	69 %	62 m/s
80 m/s	70 % (needs the governor)	63 m/s
85 m/s	70 % (needs the governor)	63 m/s
90 m/s	70 % (needs the governor)	63 m/s

SKYG

SKYG SPEED 90

